

Part II

Lattices and linear diophantine equations

Lattices and linear diophantine equations can be described, to a large extent, parallel to linear spaces and linear equations (Part I) and to polyhedra and linear inequalities (Part III).

In this part we first discuss in Chapter 4 the theoretical side of lattices and linear diophantine equations, basically due to Hermite and Minkowski. Fundamental is the *Hermite normal form* of a matrix. Next, in Chapters 5 and 6, we go into the algorithmic side.

In Chapter 5 we discuss first the classical *Euclidean algorithm* for finding the g.c.d., which also can be used for solving *one* linear diophantine equation in polynomial time. Next an extension to solving *systems* of linear diophantine equations in polynomial time is described, based on finding the Hermite normal form of a matrix.

Chapter 6 starts with a discussion of the classical *continued fraction method* for approximating a real number by rational numbers of low denominator (*diophantine approximation*). The continued fraction method is a variant of the Euclidean algorithm. Next we describe Lovász's important *basis reduction method* for lattices. This method applies to simultaneous diophantine approximation, to solving systems of linear diophantine equations, to linear programming (see Section 14.1), to integer linear programming (see Section 18.4), to factoring polynomials over $\mathbb{Q}$ in polynomial time, and to several other problems.

Part II concludes with historical and further notes on lattices and linear diophantine equations.

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4

Theory of lattices and linear diophantine equations

In this chapter we describe some results from elementary number theory concerning lattices and linear diophantine equations. In Section 4.1 we show that each rational matrix of full row rank can be brought into the so-called *Hermite normal form*. We derive a characterization for the feasibility of a system of linear diophantine equations and derive the existence of a linearly independent basis for any rational lattice. In Section 4.2 we show uniqueness of the Hermite normal form. In Section 4.3 we study *unimodular* matrices, and in Section 4.4 we go into some further theoretical aspects.

4.1. THE HERMITE NORMAL FORM

A matrix of full row rank is said to be in *Hermite normal form* if it has the form $[B \ 0]$, where B is a nonsingular, lower triangular, nonnegative matrix, in which each row has a unique maximum entry, which is located on the main diagonal of B .

The following operations on a matrix are called *elementary (unimodular) column operations*:

- (1) (i) exchanging two columns;
- (ii) multiplying a column by -1 ;
- (iii) adding an integral multiple of one column to another column.

Theorem 4.1 (Hermite normal form theorem). *Each rational matrix of full row rank can be brought into Hermite normal form by a series of elementary column operations.*

Proof. Let A be a rational matrix of full row rank. Without loss of generality, A is integral. Suppose we have transformed A , by elementary column operations, to the form $\begin{bmatrix} B & 0 \\ C & D \end{bmatrix}$ where B is lower triangular and with positive diagonal.

Now with elementary column operations we can modify D so that its first row $(\delta_{11}, \dots, \delta_{1k})$ is nonnegative, and so that the sum $\delta_{11} + \dots + \delta_{1k}$ is as small as possible. We may assume that $\delta_{11} \geq \delta_{12} \geq \dots \geq \delta_{1k}$. Then $\delta_{11} > 0$, as A has full row rank. Moreover, if $\delta_{12} > 0$, by subtracting the second column of D from the first column of D , the first row will have smaller sum, contradicting our assumption. Hence $\delta_{12} = \dots = \delta_{1k} = 0$, and we have obtained a larger lower triangular matrix (this last procedure corresponds to the Euclidean algorithm cf. Section 5.1).

By repeating this procedure, the matrix A finally will be transformed into $[B \ 0]$ with $B = (\beta_{ij})$ lower triangular with positive diagonal. Next do the following:

- (2) for $i = 2, \dots, n$ ($n :=$ order of B), do the following: for $j = 1, \dots, i - 1$, add an integer multiple of the i th column of B to the j th column of B so that the (i, j) th entry of B will be nonnegative and less than β_{ii} .

(So the procedure is applied in the order: $(i, j) = (2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3), \dots$). It is easy to see that after these elementary column operations the matrix is in Hermite normal form. $\square$

In Theorem 4.2 below we shall see that any rational matrix of full row rank has a *unique* Hermite normal form. So we can speak of *the* Hermite normal form of a matrix.

A first corollary gives necessary and sufficient conditions for the feasibility of systems of linear diophantine equations. This result is a corollary of *Kronecker's approximation theorem* [1884b] (cf. Koksma [1936: p. 83] and the historical notes at the end of Part II).

Corollary 4.1a. *Let A be a rational matrix and let b be a rational column vector. Then the system $Ax = b$ has an integral solution x , if and only if yb is an integer for each rational row vector y for which yA is integral.*

Proof. Necessity of the condition is trivial: if x and yA are integral vectors and $Ax = b$, then $yb = yAx$ is an integer.

To see sufficiency, suppose yb is an integer whenever yA is integral. Then $Ax = b$ has a (possibly fractional) solution, since otherwise $yA = 0$ and $yb = \frac{1}{2}$ for some rational vector y (by linear algebra – see Corollary 3.1b). So we may assume that the rows of A are linearly independent. Now both sides of the equivalence to be proved are invariant under elementary column operations (1). So by Theorem 4.1 we may assume that A is in Hermite normal form $[B \ 0]$. Since $B^{-1}[B \ 0] = [I \ 0]$ is an integral matrix, it follows from our assumption that also $B^{-1}b$ is an integral vector. Since

$$(3) \quad [B \ 0] \begin{pmatrix} B^{-1}b \\ 0 \end{pmatrix} = b$$

the vector $x := \begin{pmatrix} B^{-1}b \\ 0 \end{pmatrix}$ is an integral solution for $Ax = b$. $\square$

Corollary 4.1a can be seen as the integer analogue of Corollary 3.1b and of Corollary 7.1d (Farkas' lemma). It can also be interpreted in terms of groups and lattices. A subset Λ of $\mathbb{R}^n$ is called an (*additive*) *group* if:

- (4) (i) $0 \in \Lambda$
 (ii) if $x, y \in \Lambda$ then $x + y \in \Lambda$ and $-x \in \Lambda$.

The group is said to be *generated* by $a_1, \dots, a_m$ if

- (5) $\Lambda = \{\lambda_1 a_1 + \dots + \lambda_m a_m \mid \lambda_1, \dots, \lambda_m \in \mathbb{Z}\}$.

The group is called a *lattice* if it can be generated by linearly independent vectors. These vectors then are called a *basis* for the lattice.

Note that if matrix B comes from matrix A by elementary column operations, then the columns of B generate the same group as the columns of A .

Theorem 4.1 has also the following consequence.

Corollary 4.1b. *If $a_1, \dots, a_m$ are rational vectors, then the group generated by $a_1, \dots, a_m$ is a lattice, i.e. is generated by linearly independent vectors.*

Proof. We may assume that $a_1, \dots, a_m$ span all space. (Otherwise we could apply a linear transformation to a lower dimensional space.) Let A be the matrix with columns $a_1, \dots, a_m$ (so A has full row rank). Let $[B \ 0]$ be the Hermite normal form of A . Then the columns of B are linearly independent vectors generating the same group as $a_1, \dots, a_m$. $\square$

So if $a_1, \dots, a_m$ are rational vectors we can speak of the lattice generated by $a_1, \dots, a_m$. Given a rational matrix A , Corollary 4.1a gives necessary and sufficient conditions for being an element of the lattice Λ generated by the columns of A . The proof of Corollary 4.1a implies that if A has full row rank, with Hermite normal form $[B \ 0]$ (B lower triangular), then b belongs to Λ if and only if $B^{-1}b$ is integral.

Another consequence of Theorem 4.1 is:

Corollary 4.1c. *Let A be an integral $m \times n$ -matrix of full row rank. Then the following are equivalent:*

- (i) the g.c.d. of the subdeterminants of A of order m is 1;
 (ii) the system $Ax = b$ has an integral solution x , for each integral vector b ;
 (iii) for each vector y , if yA is integral, then y is integral.

Proof. Since (i), (ii), and (iii) are invariant under elementary column operations on A , by Theorem 4.1 we may assume that A is in Hermite normal form $[B \ 0]$, with B triangular. Then each of (i), (ii), and (iii) is easily seen to be equivalent to: $B = I$. $\square$

It can be derived easily from the Hermite normal form theorem that for any rational system $Ax = b$ with at least one integral solution there exist integral

vectors $x_0, x_1, \dots, x_t$ such that

$$(6) \quad \{x | Ax = b; x \text{ integral}\} = \{x_0 + \lambda_1 x_1 + \dots + \lambda_t x_t | \lambda_1, \dots, \lambda_t \in \mathbb{Z}\}$$

where $x_1, \dots, x_t$ are linearly independent and $t = (\text{number of columns of } A) - \text{rank}(A)$. The existence of such a system of fundamental solutions was stated by Smith [1861].

4.2. UNIQUENESS OF THE HERMITE NORMAL FORM

The uniqueness of the Hermite normal form follows from the following theorem.

Theorem 4.2. *Let A and A' be rational matrices of full row rank, with Hermite normal forms $[B \ 0]$ and $[B' \ 0]$, respectively. Then the columns of A generate the same lattice as those of A' , if and only if $B = B'$.*

Proof. Sufficiency is clear, as the columns of B and A generate the same lattice, and similarly for B' and A' .

To see necessity, suppose the columns of A and those of A' generate the same lattice Λ . Then the same holds for B and B' , as they come by elementary column operations from A and A' . Write $B = (\beta_{ij})$ and $B' = (\beta'_{ij})$. Suppose $B \neq B'$, and choose $\beta_{ij} \neq \beta'_{ij}$ with i as small as possible. Without loss of generality, $\beta_{ii} \geq \beta'_{ii}$. Let b_j and b'_j be the j th columns of B and B' . Then $b_j \in \Lambda$ and $b'_j \in \Lambda$, and hence $b_j - b'_j \in \Lambda$. This implies that $b_j - b'_j$ is an integral linear combination of the columns of B . By our choice of i , the vector $b_j - b'_j$ has zeros in the first $i - 1$ positions. Hence, as B is lower triangular, $b_j - b'_j$ is an integral linear combination of the columns indexed $i, \dots, n$. So $\beta_{ij} - \beta'_{ij}$ is an integral multiple of β_{ii} . However, this contradicts the fact that $0 < |\beta_{ij} - \beta'_{ij}| < \beta_{ii}$ (since if $j = i$, then $0 < \beta'_{ii} < \beta_{ii}$, and if $j < i$, then $0 \leq \beta_{ij} < \beta_{ii}$ and $0 < \beta'_{ij} < \beta'_{ii} \leq \beta_{ii}$). $\square$

Corollary 4.2a. *Every rational matrix of full row rank has a unique Hermite normal form.*

Proof. Take $A = A'$ in Theorem 4.2. $\square$

Note that if $\beta_{11}, \dots, \beta_{mm}$ are the diagonal entries of the Hermite normal form $[B \ 0]$ of A , then for each $j = 1, \dots, m$, the product $\beta_{11} \cdots \beta_{jj}$ is equal to the g.c.d. of the subdeterminants of order j of the first j rows of A (as this g.c.d. is invariant under elementary column operations). This is an alternative way of seeing that the main diagonal of the Hermite normal form is unique. (It also implies that the size of the Hermite normal form is polynomially bounded by the size of the original matrix—cf. Section 5.2.)

4.3. UNIMODULAR MATRICES

Series of elementary column operations can be described by so-called unimodular matrices. Let U be a nonsingular matrix. Then U is called *unimodular*

if U is integral and has determinant ± 1 . (Later we shall extend the notion of unimodularity to not-necessarily nonsingular matrices.) Unimodular matrices were studied by Smith [1861], Frobenius [1879, 1880], and Veblen and Franklin [1921–2].

Theorem 4.3. *The following are equivalent for a nonsingular rational matrix U of order n :*

- (i) U is unimodular;
- (ii) U^{-1} is unimodular;
- (iii) the lattice generated by the columns of U is $\mathbb{Z}^n$;
- (iv) U has the identity matrix as its Hermite normal form;
- (v) U comes from the identity matrix by elementary column operations.

Proof. (i) implies (ii), as $\det(U^{-1}) = (\det U)^{-1} = \pm 1$, and as each entry of U^{-1} is a subdeterminant of U , and hence is an integer. Similarly, (ii) implies (i).

The equivalence of (iii), (iv), and (v) follows directly from Theorem 4.2. The implication (v) $\Rightarrow$ (i) is trivial: elementary column operations leave a matrix integral, and do not change its determinant (up to sign). Also the implication (i) $\Rightarrow$ (iv) is easy: if B is the Hermite normal form of U , then B is integral and $|\det B| = |\det U| = 1$. Hence $B = I$. $\square$

Corollary 4.3a. *Let A and A' be nonsingular matrices. Then the following are equivalent:*

- (i) the columns of A and those of A' generate the same lattice;
- (ii) A' comes from A by elementary column operations;
- (iii) $A' = AU$ for some unimodular matrix U (i.e. $A^{-1}A'$ is unimodular).

Proof. Directly from Theorem 4.2 and 4.3. $\square$

Corollary 4.3b. *For each rational matrix A of full row rank there is a unimodular matrix U such that AU is the Hermite normal form of A . If A is nonsingular, U is unique.*

Proof. Directly from Corollary 4.2a and Theorem 4.3. $\square$

If A and B are nonsingular matrices, and each column of B is in the lattice generated by the columns of A , then $\det B$ is an integral multiple of $\det A$. Furthermore, $|\det A| = |\det B|$ if and only if the lattice generated by the columns of A coincides with the lattice generated by the columns of B .

Proof. If each column of B is in the lattice generated by the columns of A then $B = AU$ for some integral matrix U . Therefore, $\det B = \det U \cdot \det A$ ($\det U$ is an integer), and $|\det B| = |\det A|$ if and only if U is unimodular. $\square$

In Sections 19.1 and 21.4 we extend the notion of unimodularity to not-necessarily nonsingular matrices. Moreover, in Chapters 19–21 we shall consider

totally unimodular matrices, in which by definition all subdeterminants are 0, +1, or -1.

4.4. FURTHER REMARKS

Dual lattices

If Λ is a full-dimensional lattice (i.e. Λ spans all space), define the *dual lattice* $\Lambda^\perp$ by:

$$(7) \quad \Lambda^\perp := \{z \mid zx \text{ is an integer for all } x \in \Lambda\}.$$

If Λ is generated by the columns of the nonsingular matrix B , then $\Lambda^\perp$ is the lattice generated by the rows of B^{-1} .

[Indeed, the lattice generated by the rows of B^{-1} is contained in $\Lambda^\perp$ as each row of B^{-1} is contained in $\Lambda^\perp$ (since $B^{-1}B = I$). Conversely, if $z \in \Lambda^\perp$, then zB is integral. Hence $z = (zB)B^{-1}$ is an integral combination of the rows of B^{-1} .]

So $\Lambda^{\perp\perp} = \Lambda$.

It may be derived more generally from Theorem 4.1 that for all rational matrices A and B there are rational matrices C and D such that:

$$(8) \quad \{x \mid \exists y, \exists \text{integral } z: x = Ay + Bz\} = \{x \mid Cx = 0; Dx \text{ is integral}\}.$$

Moreover, if A, B, C, D are related as in (8) then also:

$$(9) \quad \{u \mid \exists v, \exists \text{integral } w: u = vC + wD\} = \{u \mid uA = 0; uB \text{ is integral}\}.$$

Smith normal form

Smith [1861] proved that any rational matrix A can be transformed by elementary row and column operations into the *Smith normal form* $\begin{bmatrix} D & 0 \\ 0 & 0 \end{bmatrix}$, where $D = \text{diag}(\delta_1, \dots, \delta_k)$,

with $\delta_1, \dots, \delta_k$ positive rationals such that $\delta_1 \mid \delta_2 \mid \dots \mid \delta_k$. (Elementary row operations are of course the operations (1) with column replaced by row.) This form is unique, as one easily checks that for each i , the product $\delta_1 \dots \delta_i$ is equal to the g.c.d. of the subdeterminants of A of order i (as this g.c.d. is invariant under elementary row and column operations). The numbers $\delta_1, \dots, \delta_k$ are called the *elementary divisors* of A .

To see that any rational $m \times n$ matrix can be brought into Smith normal form, suppose we have applied the elementary operations so that we obtained a matrix $B = (\beta_{ij})$ for which

$$(10) \quad \min \{ |\beta_{ij}| \mid i = 1, \dots, m; j = 1, \dots, n; \beta_{ij} \neq 0 \}$$

is as small as possible. We may assume this minimum is attained by β_{11} . Moreover, we may assume that all other entries in the first row and column are 0, as by applying elementary operations we can have $0 \leq \beta_{1j} < |\beta_{11}|$ for all $j \neq 1$ and $0 \leq \beta_{i1} < |\beta_{11}|$ for all $i \neq 1$. Hence, by the minimality of $|\beta_{11}|$, $\beta_{1j} = 0$ and $\beta_{i1} = 0$ for $j \neq 1$ and $i \neq 1$. Now each entry of B is an integral multiple of β_{11} . Indeed, if β_{ij} is not a multiple of β_{11} , with elementary operations we can obtain the remainder of $\beta_{ij} \pmod{\beta_{11}}$ in the (i, j) th position, contradicting again the minimality of $|\beta_{11}|$. Write

$$(11) \quad B = \begin{bmatrix} \beta_{11} & 0 \\ 0 & C \end{bmatrix}.$$

By induction, C can be transformed by elementary operations into Smith normal form

$\begin{bmatrix} D' & 0 \\ 0 & 0 \end{bmatrix}$, say with $D' = \text{diag}(\delta_2, \dots, \delta_k)$ and $\delta_2 | \delta_3 | \dots | \delta_k$. Since δ_{11} divides all entries of C , β_{11} also divides δ_2 , and hence also B can be brought into Smith normal form.

As for each j , the product $\delta_1 \cdots \delta_j$ is equal to the g.c.d. of the subdeterminants of A of order j , the size of the Smith normal form is polynomially bounded by the size of A . By the equivalence of (i) and (v) in Theorem 4.3, for any matrix A there are unimodular matrices U and V such that UAV is in Smith normal form.

A consequence of the existence of the Smith normal form is the famous theorem of Kronecker [1877] that each finite abelian group is the direct product of cyclic groups (see Cohn [1962]). The Smith normal form was also studied by Frobenius [1879, 1880].

It can be proved that if A and B are rational nonsingular matrices of order n , and $\text{g.c.d.}(\det A, \det B) = 1$, then $\text{Snf}(A) \cdot \text{Snf}(B) = \text{Snf}(AB)$, where Snf denotes Smith normal form (see Newman [1972: pp. 33–34] and Marcus and Underwood [1972]).

Other characterizations of feasibility of linear diophantine equations

Corollary 4.1a extends the well-known result from elementary number theory that one linear equation $x_1 \xi_1 + \dots + x_n \xi_n = \beta$ has a solution in integers $\xi_1, \dots, \xi_n$, if and only if β is an integral multiple of the g.c.d. of $x_1, \dots, x_n$. Another extension of this result to more dimensions is a characterization due to Heger [1858: p. 111] (cf. Smith [1861], Frobenius [1879], Veblen and Franklin [1921], Skolem [1938]):

- (12) *let A be a rational matrix of full row rank, with m rows, and let b be a rational column m -vector. Then $Ax = b$ has an integral solution x , if and only if each subdeterminant of the matrix $[A \ b]$ of order m is an integral multiple of the g.c.d. of the subdeterminants of the matrix A of order m .*

This may be proved in a similar way as Corollary 4.1a: both sides of the equivalence are invariant under elementary column operations on A , and the equivalence is easy for matrices in Hermite normal form.

Another characterization is:

- (13) *the rational system $Ax = b$ has an integral solution x if and only if for each natural number M the congruence $Ax \equiv b \pmod{M}$ has an integral solution x*

(cf. Mordell [1969]). This may be derived from Corollary 4.1a as follows. The necessity of the condition is trivial. To prove sufficiency, suppose $Ax = b$ has no integral solution. Then by Corollary 4.1a there exists a rational vector y with yA integral and yb not an integer. Let M be a natural number such that My is integral. Let x be an integral solution for $Ax \equiv b \pmod{M}$. So $M^{-1}(Ax - b)$ is integral. Hence $y(Ax - b)$ is an integer. Since yAx is an integer, it follows that yb is an integer. Contradiction.

For studies of the duality aspects of Corollary 4.1a, see Bachem and von Radow [1979] and Williams [1984].

5

Algorithms for linear diophantine equations

The classical Euclidean algorithm for finding the g.c.d. yields also a polynomial-time method for solving one rational linear diophantine equation. This is described in Section 5.1, as a preliminary to solving *systems* of rational linear diophantine equations, discussed in Sections 5.2 and 5.3. First in Section 5.2 we show that the size of the Hermite normal form of a rational matrix A is polynomially bounded by the size of A , and we derive that the solvability of systems of rational linear diophantine equations has a good characterization. Next in Section 5.3 we describe a polynomial-time algorithm for finding the Hermite normal form of a matrix, and we deduce that systems of rational linear diophantine equations can be solved in polynomial time.

5.1. THE EUCLIDEAN ALGORITHM

The Euclidean algorithm determines, in polynomial time, the g.c.d. of two positive rational numbers α and β . First α is replaced by $\alpha - \lfloor \alpha/\beta \rfloor \beta$, and (with the new α) β is replaced by $\beta - \lfloor \beta/\alpha \rfloor \alpha$. Next this is repeated until one of α and β is 0. In that case, the nonzero among α and β is the g.c.d. of the original α and β . This follows easily from the facts that $\text{g.c.d. } \{\alpha, \beta\} = \text{g.c.d. } \{\alpha - \lfloor \alpha/\beta \rfloor \beta, \beta\}$ and $\text{g.c.d. } \{\alpha, 0\} = \alpha$.

With a little more administration one can find integers γ and ε satisfying $\gamma\alpha + \varepsilon\beta = \text{g.c.d. } \{\alpha, \beta\}$, given rational numbers α and β , and one can solve one rational linear diophantine equation, as we shall describe now.

Let two positive rationals α and β be given. Determine a series of 3×2 -matrices $A_0, A_1, A_2, \dots$ as follows. The matrix A_0 is given by:

$$(1) \quad A_0 := \begin{bmatrix} \alpha & \beta \\ 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

If A_k has been found, say:

$$(2) \quad A_k = \begin{bmatrix} \alpha_k & \beta_k \\ \gamma_k & \delta_k \\ \varepsilon_k & \zeta_k \end{bmatrix},$$

let A_{k+1} arise from A_k by the following rule:

- (3) (i) if k is even and $\beta_k > 0$, subtract $\lfloor \alpha_k/\beta_k \rfloor$ times the second column of A_k from the first column;
 (ii) if k is odd and $\alpha_k > 0$, subtract $\lfloor \beta_k/\alpha_k \rfloor$ times the first column of A_k from the second column;

This step is performed for $k = 0, 1, 2, \dots$, until for $k = N$ (say) $\alpha_N = 0$ or $\beta_N = 0$. If $\beta_N = 0$ and $\alpha_N \neq 0$, then α_N is the g.c.d. of α and β , as during the iterations (3) the g.c.d. of the entries in the upper row of A_k does not change, and as $\alpha_N = \text{g.c.d.} \{ \alpha_N, 0 \}$. Similarly, if $\alpha_N = 0$, $\beta_N \neq 0$, then $\beta_N = \text{g.c.d.} \{ \alpha, \beta \}$.

We also find integers γ and ε with $\gamma\alpha + \varepsilon\beta = \text{g.c.d.} \{ \alpha, \beta \}$. To see this, observe that $(1, -\alpha, -\beta)A_0 = (0, 0)$, and hence for each k , $(1, -\alpha, -\beta)A_k = (0, 0)$ as we did only elementary column operations. In the notation (2) this means:

$$(4) \quad \begin{aligned} \gamma_k\alpha + \varepsilon_k\beta &= \alpha_k \\ \delta_k\alpha + \zeta_k\beta &= \beta_k. \end{aligned}$$

Therefore, if $\beta_N = 0$, $\alpha_N \neq 0$,

$$(5) \quad \begin{aligned} \gamma_N\alpha + \varepsilon_N\beta &= \alpha_N = \text{g.c.d.} \{ \alpha, \beta \} \\ \delta_N\alpha + \zeta_N\beta &= 0. \end{aligned}$$

Similarly, if $\alpha_N = 0$, $\beta_N \neq 0$.

The Euclidean algorithm indeed terminates, within polynomial time. Before proving this, we observe that for each k :

$$(6) \quad \begin{aligned} -\alpha_k\delta_k + \beta_k\gamma_k &= \beta \\ \alpha_k\zeta_k - \beta_k\varepsilon_k &= \alpha. \end{aligned}$$

The first line follows from the fact that the upper 2×2 -determinants of A_0 and A_k are the same (as we did only elementary column operations), and the second line follows similarly by leaving out the middle rows of A_0 and A_k . By considering the lower 2×2 -determinants of A_0 and A_k we obtain the equality:

$$(7) \quad \gamma_k\zeta_k - \delta_k\varepsilon_k = 1.$$

Note moreover that for all k , $\gamma_k \geq 1$, $\zeta_k \geq 1$, and $\delta_k \leq 0$, $\varepsilon_k \leq 0$.

Theorem 5.1. *The Euclidean algorithm is a polynomial-time method.*

Proof. We may assume that α and β are natural numbers (otherwise, multiply throughout the procedure the first row of each matrix A_k by the product of the denominators of α and β ; this modification does not change the complexity).

So all matrices A_k have nonnegative integers in their first row. Since at each iteration either α_k or β_k is reduced by a factor of at least 2, after at most $\lfloor \log_2 \alpha \rfloor + \lfloor \log_2 \beta \rfloor + 1$ iterations, one of α_k and β_k will be 0.

Now for each k : $\alpha_k \leq \alpha$ and $\beta_k \leq \beta$. From (6) it follows that $|\delta_k| \leq \beta$, $|\gamma_k| \leq \beta$, $|\zeta_k| \leq \alpha$, $|\varepsilon_k| \leq \alpha$, as long as $\alpha_k > 0$ and $\beta_k > 0$. This shows that throughout the method the sizes of the numbers are polynomially bounded by the sizes of α and β . $\square$

Corollary 5.1a. *A linear diophantine equation with rational coefficients can be solved in polynomial time.*

Proof. Let

$$(8) \quad \alpha_1 \xi_1 + \cdots + \alpha_n \xi_n = \beta$$

be a rational linear diophantine equation. The algorithm solving (8) is described recursively on n . The case $n = 1$ is trivial. Let $n \geq 2$. Find, with the Euclidean algorithm, α' , γ , and ε satisfying

$$(9) \quad \alpha' = \text{g.c.d.} \{ \alpha_1, \alpha_2 \} = \alpha_1 \gamma + \alpha_2 \varepsilon$$

γ, ε integers.

Next solve the linear diophantine equation (in $n - 1$ variables)

$$(10) \quad \alpha' \xi' + \alpha_3 \xi_3 + \cdots + \alpha_n \xi_n = \beta.$$

If (10) has no integral solution, then neither has (8). If (10) has an integral solution $\xi', \xi_3, \dots, \xi_n$, then $\xi_1 := \gamma \xi'$, $\xi_2 := \varepsilon \xi'$, $\xi_3, \dots, \xi_n$ gives an integral solution to (8). This defines a polynomial algorithm. $\square$

In Section 5.3 we shall see that also *systems* of linear diophantine equations can be solved in polynomial time.

Notes for Section 5.1. The fact that the number of iterations in the Euclidean algorithm is bounded linearly in the logarithm of the smallest number was proved by Lamé [1844]. For more on the Euclidean algorithm, see Knuth [1969: pp.316–364], Blankinship [1963], Bradley [1970a], Collins [1974], Dixon [1970, 1971], Harris [1970a], Heilbronn [1968], Rieger [1976], Schönhage [1971], and van Trigt [1978]. Heilbronn [1968] and Dixon [1970] showed that the number of iterations in the Euclidean algorithm applied to α, β ($\alpha \leq \beta$) is 'almost always about' $12\pi^{-2} \log 2 \cdot \log \beta$. See also Harris [1970b], Merkes and Meyers [1973], Shea [1973], and Yao and Knuth [1975].

The solution of one linear diophantine equation was studied by Bond [1967], Kertzner [1981], Lehmer [1919], Levit [1956], Morito and Salkin [1979, 1980], and Weinstock [1960].

5.2. SIZES AND GOOD CHARACTERIZATIONS

We show that the Hermite normal form has polynomially bounded size, from which it follows that the characterizations given in Section 4.1 are good characterizations.

Theorem 5.2. *The Hermite normal form $[B \ 0]$ of a rational matrix A of full row rank has size polynomially bounded by the size of A . Moreover, there exists a unimodular matrix U with $AU = [B \ 0]$, such that the size of U is polynomially bounded by the size of A .*

Proof. We may assume that A is integral, as multiplying A by the product, say κ , of the denominators occurring in A also multiplies the Hermite normal form of A by κ .

The diagonal entries of B are divisors of subdeterminants of A (cf. Section 4.2). As each row of B has its maximum entry on the main diagonal of B , it follows that the size of $[B \ 0]$ is polynomially bounded by the size of A .

The second statement can be seen as follows. By permuting columns we may assume that $A = [A' \ A'']$, with A' nonsingular. The Hermite normal form of the matrix

$$(11) \quad \begin{bmatrix} A' & A'' \\ 0 & I \end{bmatrix} \text{ is } \begin{bmatrix} B & 0 \\ B' & B'' \end{bmatrix}$$

for certain matrices B' and B'' . As the sizes of B , B' , and B'' are polynomially bounded by the size of A , also the size of the unimodular matrix

$$(12) \quad U := \begin{bmatrix} A' & A'' \\ 0 & I \end{bmatrix}^{-1} \begin{bmatrix} B & 0 \\ B' & B'' \end{bmatrix}$$

is polynomially bounded by the size of A . Now $AU = [A' \ A'']U = [B \ 0]$. $\square$

Corollary 5.2a. *If a rational system $Ax = b$ has an integral solution, it has one of size polynomially bounded by the sizes of A and b .*

Proof. Without loss of generality, A has full row rank. Let $[B \ 0] = AU$ be the Hermite normal form of A , with U unimodular of size polynomially bounded by the size of A . Now $B^{-1}b$ is integral (as for any integral solution x of $Ax = b$ we have that $B^{-1}b = B^{-1}Ax = B^{-1}[B \ 0]U^{-1}x = [I \ 0]U^{-1}x$ is integral, since $U^{-1}x$ is integral). Therefore,

$$(13) \quad \tilde{x} := U \begin{pmatrix} B^{-1}b \\ 0 \end{pmatrix}$$

is an integral solution of $Ax = b$, of size polynomially bounded by the sizes of A and b . $\square$

This implies that Corollary 4.1a in fact yields a good characterization (cf. von zur Gathen and Sieveking [1976, 1978]).

Corollary 5.2b. *The following problem has a good characterization: given a rational matrix A and a rational vector b , does the system $Ax = b$ have an integral solution?*

Proof. The general case can be easily reduced to the case where A has full row rank (using Corollary 3.3a). Let $[B \ 0] = AU$ be the Hermite normal form of A , with U unimodular of size polynomially bounded by the size of A .

If the answer is positive, there is a solution of polynomial size, by Corollary 5.2a.

If the answer is negative, by Corollary 4.1a there exists a rational row vector y with yA integral and yb not an integer. We can take y of polynomially bounded size: since $B^{-1}A$ is integral but $B^{-1}b$ not, we can take for y one of the rows of B^{-1} . $\square$

5.3. POLYNOMIAL ALGORITHMS FOR HERMITE NORMAL FORMS AND SYSTEMS OF LINEAR DIOPHANTINE EQUATIONS

The Euclidean algorithm gives the polynomial solvability of one linear diophantine equation—see Corollary 5.1a. It was shown by Frumkin [1976a, b] and von zur Gathen and Sieveking [1976] that also *systems* of linear diophantine equations can be solved in polynomial time.

Moreover, von zur Gathen and Sieveking [1976] and Votyakov and Frumkin [1976] showed that for any system $Ax = 0$ of homogeneous linear diophantine equations we can find a basis for the solution set in polynomial time. Frumkin [1976c] derived from this that also the Hermite normal form of a matrix can be determined in polynomial time. (Frumkin [1975, 1977] announces and reviews his and Votyakov's results.) A direct polynomial-time method bringing a matrix into Hermite normal form was given by Kannan and Bachem [1979]. They also gave a polynomial algorithm for the Smith normal form.

Several prescriptions for the order of the elementary column operations making the Hermite normal form had been proposed before (cf. Pace and Barnett [1974], Blankinship [1966a, b], Bodewig [1956], Bradley [1971a], Cheema [1966], Donaldson [1979], and Hu [1969: Appendix A]), but none of these methods could be shown to be polynomial-time.

The methods of Frumkin, Votyakov, von zur Gathen and Sieveking are based on an elimination process, at the same time reducing all numbers modulo a certain integer. Here we describe a variant of these methods (developed independently by Domich [1983] and Domich, Kannan and Trotter [1985]), which finds the Hermite normal form in polynomial time.

Polynomial algorithm to determine the Hermite normal form

Let an integral $m \times n$ -matrix A , of full row rank, be given. Let M be the absolute value of the determinant of an (arbitrary) submatrix of A of rank m (such an M can be found in polynomial time with Gaussian elimination). Now the columns of A generate the same lattice as the columns of the matrix:

$$(14) \quad A' := \left[A \mid \begin{array}{c} M \\ \vdots \\ 0 \\ \vdots \\ 0 \\ \vdots \\ M \end{array} \right]$$

[Proof. A has a submatrix B of rank m with $|\det B| = M$. Then the matrix $(\det B)B^{-1}$ is integral, and $B((\det B)B^{-1}) = \pm M \cdot I$, so that each of the last m columns of A' is an integral combination of the columns of B . $\square$]

So the Hermite normal form of A is the same as that of A' , except for the last m columns of the Hermite normal form of A' , which are all-zero.

So it suffices to find the Hermite normal form of A' . The idea is that after each elementary column operation to bring A' to Hermite normal form we can do some further elementary column operations to bring all entries between 0 and M . More precisely, first we add integral multiples of the last m columns of A' to the first n columns of A' so that all components will be at least 0 and at most M . Next we apply the following procedure repeatedly:

Suppose we have obtained the matrix

$$(15) \quad \left[\begin{array}{c|c|c} B & 0 & 0 \\ \hline & & 0 \dots 0 \\ C & D & \begin{matrix} M & & 0 \\ & \ddots & \\ 0 & & M \end{matrix} \end{array} \right]$$

where B is a lower triangular $k \times k$ -matrix, C is an $(m-k) \times k$ -matrix, D is an $(m-k) \times (n+1)$ -matrix, such that the first row of D is nonzero. (In particular, (14) has form (15), with $k=0$.)

Then apply the following repeatedly, writing $D = (\delta_{ij})_{i=1, j=1}^{m-k, n+1}$:

- (16) if there are $i \neq j$ with $\delta_{1i} \geq \delta_{1j} > 0$, then:
- (i) subtract $\lfloor \delta_{1i}/\delta_{1j} \rfloor$ times the j th column of D from the i th column of D ;
 - (ii) next add integral multiples of the last $m-k-1$ columns of (15) to the other columns to bring all components between 0 and M (i.e. reduce D modulo M).

(16) is repeated as long as the first row of D contains more than one nonzero entry. If it has exactly one nonzero, we start the procedure anew, with k replaced by $k+1$.

If $k=m$, (15) is in the form $[B \ 0]$ with B lower triangular. Then finally with the following procedure it can be brought into Hermite normal form:

- (17) for $i=2, \dots, n$ ($:=$ order of B), do the following: for $j=1, \dots, i-1$, add an integral multiple of the i th column of B to the j th column of B so that the (i, j) th entry of B will be nonnegative and less than β_{ii} .

Deleting the last m columns yields the Hermite normal form of A .

Theorem 5.3. *The above method finds the Hermite normal form in polynomial time.*

Proof. The execution time of each single application of (i) or (ii) in (16) can be easily seen to be polynomially bounded by n and $\log_2 M$. Since at each application of step (16), either the first row of D will contain one zero entry more,

or the product of the nonzero entries in this row is reduced by a factor of at least 2 (as $(\delta_{1i} - \lfloor \delta_{1i}/\delta_{1j} \rfloor \delta_{1j}) \leq \frac{1}{2} \delta_{1i}$), and since all entries in the first row of D are at most M , after at most $n \cdot \log_2 M$ applications of step (16) the matrix D has at most one nonzero in the first row. In that case k is increased to $k + 1$.

So after at most $mn \cdot \log_2 M$ applications of (16) we have $k = m$. It is easy to see that during application of the final procedure (17) the numbers will not be larger than M^n in absolute value. Therefore, the running time of the algorithm is polynomially bounded. $\square$

Clearly, also, if A is not integral the Hermite normal form can be found in polynomial time: either we apply the above method, or we first multiply A by the product of the denominators of the entries of A .

Also a unimodular matrix bringing a matrix into Hermite normal form can be found in polynomial time.

Corollary 5.3a. *Given a rational matrix A of full row rank, we can find in polynomial time a unimodular matrix U such that AU is in Hermite normal form.*

Proof. Similar to the proof of Theorem 5.2. $\square$

Another consequence is a result of Frumkin [1976a, b] and von zur Gathen and Sieveking [1976].

Corollary 5.3b. *Given a system of rational linear equations, we can decide if it has an integral solution, and if so, find one, in polynomial time.*

Proof. Let the system $Ax = b$ be given. With the Gaussian elimination method we can decide if $Ax = b$ has any (possibly fractional) solution. If not, it certainly has no integral solution. If so, we can restrict $Ax = b$ to a subsystem with linearly independent left-hand sides. So we may assume that A has full row rank. Determine with the above method the Hermite normal form $[B \ 0]$ of A , and determine a unimodular matrix U with $AU = [B \ 0]$. If $B^{-1}b$ is integral, then

$$(18) \quad x := U \begin{pmatrix} B^{-1}b \\ 0 \end{pmatrix}$$

is an integral solution for $Ax = b$. If $B^{-1}b$ is not integral, $Ax = b$ has no integral solution. $\square$

Von zur Gathen and Sieveking [1976] and Votyakov and Frumkin [1976] showed:

Corollary 5.3c. *Given a feasible system $Ax = b$ of rational linear diophantine equations, we can find in polynomial time integral vectors $x_0, x_1, \dots, x_t$ such that*

$$(19) \quad \{x \mid Ax = b; x \text{ integral}\} = \{x_0 + \lambda_1 x_1 + \dots + \lambda_t x_t \mid \lambda_1, \dots, \lambda_t \in \mathbb{Z}\}$$

with $x_1, \dots, x_t$ linearly independent.

Proof. As in the proof of Corollary 5.3b we may assume that A has full row rank, and we can find in polynomial time a unimodular matrix U such that $AU = [B \ 0]$ is in Hermite normal form. Let $x_0, x_1, \dots, x_t$ be given by

$$(20) \quad [x_0, x_1, \dots, x_t] = U \begin{bmatrix} B^{-1}b & 0 \\ 0 & I \end{bmatrix}.$$

Then $x_0, x_1, \dots, x_t$ have the required properties. $\square$

Notes for Chapter 5. Further work on solving systems of linear diophantine equations is described by Chou and Collins [1982]. Also Fiorot and Gondran [1969] study solving systems of linear diophantine equations. Rosser [1942] studied extensions of the Euclidean algorithm to more than two variables. See also the references to extensions of the continued fraction method to more dimensions, given at the end of Chapter 6.

Computing the *Smith normal form* was studied by Bodewig [1956], Hu [1969: Appendix A], Bradley [1971a], Garfinkel and Nemhauser [1972a: Sec. 7.4], Pace and Barnett [1974], Hartley and Hawkes [1974], Rayward-Smith [1979], and Kannan and Bachem [1979]. The last reference gives a polynomial-time algorithm (see also Bachem and Kannan [1979] (applications)).

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6

Diophantine approximation and basis reduction

We now study the problems of approximating real numbers by rational numbers of low denominator, i.e. *diophantine approximation*, and of finding a so-called *reduced basis* in a lattice. First, in Section 6.1, we discuss the *continued fraction method* for approximating one real number. Next, in Section 6.2, we describe Lovász's *basis reduction method* for lattices, finding a 'reduced' basis in a lattice. In Section 6.3 we give some of the several applications of this method, such as finding a short nonzero vector in a lattice and *simultaneous diophantine approximation* (approximating several reals by rationals with one common low denominator).

6.1. THE CONTINUED FRACTION METHOD

The continued fraction method, which may be seen as a variant of the Euclidean algorithm, yields approximations of a real number by rationals. It can be used to find integers p and q as described in the following classical result of Dirichlet [1842]:

Theorem 6.1 (Dirichlet's theorem). *Let α be a real number and let $0 < \varepsilon \leq 1$. Then there exist integers p and q such that:*

$$(1) \quad \left| \alpha - \frac{p}{q} \right| < \frac{\varepsilon}{q} \quad \text{and} \quad 1 \leq q \leq \varepsilon^{-1}.$$

Proof. The proof is based on Dirichlet's *pigeon-hole principle*. Let $M := \lfloor \varepsilon^{-1} \rfloor$, and consider the numbers

$$(2) \quad \{0\alpha\}, \{1\alpha\}, \{2\alpha\}, \{3\alpha\}, \dots, \{M\alpha\}$$

where $\{ \}$ denotes fractional part (i.e. $\{\beta\} = \beta - \lfloor \beta \rfloor$). Since $0 \leq \{i\alpha\} < 1$, there exist two different integers i and j with $0 \leq i, j \leq M$ and $\{(i-j)\alpha\} \leq 1/(M+1)$.

Let $q := i - j$ and $p := [q\alpha]$. Then

$$(3) \quad \left| \alpha - \frac{p}{q} \right| = \left| \frac{\{(i-j)\alpha\}}{q} \right| \leq \frac{1}{(M+1)|q|} < \frac{\varepsilon}{|q|}.$$

As $|q| \leq M \leq \varepsilon^{-1}$, Dirichlet's result (1) follows. $\square$

The *continued fraction method* for the real number $\alpha \geq 0$ can be described as follows. Determine $\gamma_1, \gamma_2, \gamma_3, \dots$ successively by:

$$(4) \quad \begin{aligned} \gamma_1 &:= \alpha \\ \gamma_2 &:= (\gamma_1 - [\gamma_1])^{-1} \\ \gamma_3 &:= (\gamma_2 - [\gamma_2])^{-1} \\ \gamma_4 &:= (\gamma_3 - [\gamma_3])^{-1} \\ &\vdots \end{aligned}$$

The sequence stops if γ_N , say, is an integer, so that we cannot determine γ_{N+1} . Directly from (4) we see that $\gamma_1, \gamma_2, \gamma_3, \dots$ satisfy:

$$(5) \quad \alpha = \gamma_1 = [\gamma_1] + \frac{1}{\gamma_2} = [\gamma_1] + \frac{1}{[\gamma_2] + \frac{1}{\gamma_3}} = [\gamma_1] + \frac{1}{[\gamma_2] + \frac{1}{[\gamma_3] + \frac{1}{\gamma_4}}} = \dots$$

The associated rationals

$$(6) \quad [\gamma_1], \quad [\gamma_1] + \frac{1}{[\gamma_2]}, \quad [\gamma_1] + \frac{1}{[\gamma_2] + \frac{1}{[\gamma_3]}}, \\ [\gamma_1] + \frac{1}{[\gamma_2] + \frac{1}{[\gamma_3] + \frac{1}{[\gamma_4]}}}, \dots$$

are called the *convergents* for α . If this sequence stops, its last term is equal to α , as one easily sees. It will be shown below that the sequence stops iff α is rational. For irrational α , the sequence converges to α .

The convergents can be described alternatively as follows. Let $\alpha \geq 0$ be a real number. Let p_1/q_1 and p_2/q_2 be the first two terms in (6), i.e.

$$(7) \quad \frac{p_1}{q_1} = [\alpha], \quad \frac{p_2}{q_2} = [\alpha] + \frac{1}{[\alpha - [\alpha]]^{-1}}$$

where p_1, q_1, p_2, q_2 are nonnegative integers, with $\text{g.c.d.}(p_1, q_1) = \text{g.c.d.}(p_2, q_2) = 1$ (if α is by chance an integer, we only determine p_1, q_1). Note that $p_1/q_1 \leq \alpha \leq p_2/q_2$ and $p_2q_1 - p_1q_2 = 1$.

Suppose we have found nonnegative integers $p_{k-1}, q_{k-1}, p_k, q_k$ such that

$$(8) \quad \frac{p_{k-1}}{q_{k-1}} < \alpha < \frac{p_k}{q_k} \quad \text{and} \quad p_k q_{k-1} - p_{k-1} q_k = 1$$

where k is even. ((8) implies $\text{g.c.d.}(p_{k-1}, q_{k-1}) = \text{g.c.d.}(p_k, q_k) = 1$.) Let t be the largest integer such that

$$(9) \quad \frac{p_{k-1} + t p_k}{q_{k-1} + t q_k} \leq \alpha.$$

Define

$$(10) \quad \begin{aligned} p_{k+1} &:= p_{k-1} + t p_k \\ q_{k+1} &:= q_{k-1} + t q_k. \end{aligned}$$

One easily checks $p_k q_{k+1} - p_{k+1} q_k = 1$, implying $\text{g.c.d.}(p_{k+1}, q_{k+1}) = 1$.

(9) and (10) give: $p_{k+1}/q_{k+1} \leq \alpha$. In the case of equality we stop. In the case of strict inequality, let u be the largest integer such that

$$(11) \quad \alpha \leq \frac{p_k + u p_{k+1}}{q_k + u q_{k+1}}.$$

Define

$$(12) \quad \begin{aligned} p_{k+2} &:= p_k + u p_{k+1} \\ q_{k+2} &:= q_k + u q_{k+1}. \end{aligned}$$

Again, one easily checks $p_{k+2} q_{k+1} - p_{k+1} q_{k+2} = 1$, implying $\text{g.c.d.}(p_{k+2}, q_{k+2}) = 1$.

(11) and (12) give $\alpha \leq p_{k+2}/q_{k+2}$. In the case of equality we stop. In the case of strict inequality, we know that (8) holds, with k increased by 2. So we can repeat the iteration, thus defining the sequence

$$(13) \quad \frac{p_1}{q_1}, \frac{p_2}{q_2}, \frac{p_3}{q_3}, \dots$$

If the sequence stops, its last term is α . It turns out that the sequence (13) is the same as the sequence (6) of convergents.

Proof. We first prove:

Claim. If $0 < \alpha < 1$ and $p_1/q_1, p_2/q_2, p_3/q_3, \dots$ is row (13) with respect to α , then $q_2/p_2, q_3/p_3, q_4/p_4, \dots$ is row (13) with respect to $1/\alpha$.

Proof of the claim. Let $p'_1/q'_1, p'_2/q'_2, p'_3/q'_3, \dots$ be row (13) with respect to $1/\alpha$. Then the following holds:

$$(14) \quad \frac{q_2}{p_2} = \frac{p'_1}{q'_1} \leq \frac{1}{\alpha} \leq \frac{p'_2}{q'_2} = \frac{q_3}{p_3}.$$

Indeed,

$$(15) \quad \frac{p'_1}{q'_1} = \left\lfloor \frac{1}{\alpha} \right\rfloor = \frac{q_2}{p_2},$$

6.1 The continued fraction method

$$\frac{p'_2}{q'_2} = \left[\frac{1}{x} \right] + \frac{1}{\left[(1/x - \left[1/x \right])^{-1} \right]} = \left[\frac{1}{x} \right] + \frac{1}{t} = \frac{q_2}{p_2} + \frac{1}{t} = \frac{q_2 t + q_1}{p_2 t + p_1} = \frac{q_3}{p_3}$$

where t is the largest integer with

$$(16) \quad \frac{p_2 t + p_1}{q_2 t + q_1} \leq x, \quad \text{i.e.} \quad t = \left\lfloor \frac{x q_1 - p_1}{p_2 - x q_2} \right\rfloor = \left\lfloor \left(\frac{1}{x} - \left[\frac{1}{x} \right] \right)^{-1} \right\rfloor,$$

using $p_1 = 0, q_1 = 1, p_2 = 1, q_2 = \lfloor 1/x \rfloor$. This proves (14).

By considering the way (13) is constructed, (14) implies that for each even k :

$$(17) \quad \frac{q_k}{p_k} = \frac{p'_{k-1}}{q'_{k-1}} \leq \frac{1}{x} \leq \frac{p'_k}{q'_k} = \frac{q_{k+1}}{p_{k+1}}.$$

End of proof of the claim.

We next show that (13) is the same sequence as (6). Let $c_k(x)$ denote the k th convergent of x , i.e. the k th term in (6). We show that $c_k(x) = p_k/q_k$ by induction on k . We may assume that $0 < x < 1$, since each term both in (6) and in (13) decreases by $\lfloor x \rfloor$ if we replace x by $x - \lfloor x \rfloor$, and since the case $x = 0$ is trivial.

If $0 < x < 1$, then $c_k(x) = (c_{k-1}(1/x))^{-1}$, as one easily derives from (6). By induction and by the Claim, $c_{k-1}(1/x) = q_k/p_k$. Hence $c_k(x) = p_k/q_k$. $\square$

Note that for every k

$$(18) \quad \left| \frac{p_{k+1}}{q_{k+1}} - \frac{p_k}{q_k} \right| = \frac{1}{q_k q_{k+1}}$$

since $|p_{k+1}q_k - p_k q_{k+1}| = 1$. Moreover, the denominators $q_1, q_2, q_3, \dots$ are monotonically increasing. So

$$(19) \quad \left| x - \frac{p_k}{q_k} \right| \leq \left| \frac{p_{k+1}}{q_{k+1}} - \frac{p_k}{q_k} \right| = \frac{1}{q_k q_{k+1}} < \frac{1}{q_k^2}.$$

Therefore by calculating the convergents we find integers p, q for Dirichlet's result (1).

Theorem 6.2. For any real numbers x and ε with $x \geq 0, 0 < \varepsilon \leq 1$, one of the convergents p_k/q_k for x satisfies (1).

Proof. Let p_k/q_k be the last convergent for which $q_k \leq \varepsilon^{-1}$ holds. If $x \neq p_k/q_k$, then p_{k+1}/q_{k+1} is defined, and by (19),

$$(20) \quad \left| x - \frac{p_k}{q_k} \right| \leq \frac{1}{q_k q_{k+1}} < \frac{\varepsilon}{q_k}.$$

If $x = p_k/q_k$, then trivially $|x - p_k/q_k| < \varepsilon/q_k$. $\square$

In fact we can derive the following. As t respectively u are the largest integers satisfying (9) resp. (11), it follows that

$$(21) \quad \frac{p_k + p_{k+1}}{q_k + q_{k+1}} > x \quad \text{if } k \text{ is even,} \quad \frac{p_k + p_{k+1}}{q_k + q_{k+1}} < x \quad \text{if } k \text{ is odd.}$$

Therefore, u resp. t (in the next iteration) are at least 1. Hence $q_{k+2} \geq q_k + q_{k+1}$ for every k . Since $q_1 = 1$ and $q_2 \geq 1$ we know inductively that $q_{k+2} \geq (\frac{3}{2})^k$ (in fact, $q_{k+2} \geq F_k$, where F_k is the k th Fibonacci number). So with (19) we know that the convergents converge to α with 'exponential convergence speed'.

If α is rational, there will be a convergent equal to α , so that the sequence of convergents stops, the last term being α .

Suppose α is rational, and the sequence does not stop. Let $\alpha = p/q$, and let p_k/q_k be the first convergent with $q_k > q$. Then we obtain the contradiction:

$$(22) \quad \frac{1}{qq_{k-1}} \leq \left| \alpha - \frac{p_{k-1}}{q_{k-1}} \right| \leq \frac{1}{q_k q_{k-1}} < \frac{1}{qq_{k-1}}.$$

Here the first inequality is based on the fact that $\alpha \neq p_{k-1}/q_{k-1}$.

Algorithmically, the convergents in the continued fraction method can be found in a way analogous to the Euclidean algorithm. The continued fraction method can be viewed as computing the g.c.d. of α and 1. For $\alpha \geq 0$, let

$$(23) \quad A_0 := \begin{bmatrix} \alpha & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

If A_k has been found, say:

$$(24) \quad A_k = \begin{bmatrix} \alpha_k & \beta_k \\ \gamma_k & \delta_k \\ \varepsilon_k & \zeta_k \end{bmatrix},$$

let A_{k+1} arise from A_k by the following rule:

- 25 (i) if k is even and $\beta_k > 0$, subtract $\lfloor \alpha_k/\beta_k \rfloor$ times the second column of A_k from the first column;
 (ii) if k is odd and $\alpha_k > 0$, subtract $\lfloor \beta_k/\alpha_k \rfloor$ times the first column of A_k from the second column.

This step is performed for $k = 0, 1, 2, \dots$, until $\alpha_k = 0$ or $\beta_k = 0$.

Similarly as for the Euclidean algorithm, we know that for each k :

$$(26) \quad \begin{aligned} \gamma_k \alpha + \varepsilon_k &= \alpha_k \\ \delta_k \alpha + \zeta_k &= \beta_k. \end{aligned}$$

Now let the matrices $A_0, A_1, A_2, \dots$ look like:

$$(27) \quad \begin{bmatrix} \alpha & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} \alpha_1 & 1 \\ q_1 & 0 \\ -p_1 & 1 \end{bmatrix}, \begin{bmatrix} \alpha_1 & \beta_2 \\ q_1 & -q_2 \\ -p_1 & p_2 \end{bmatrix}, \\ \begin{bmatrix} \alpha_3 & \beta_2 \\ q_3 & -q_2 \\ -p_3 & p_2 \end{bmatrix}, \begin{bmatrix} \alpha_3 & \beta_4 \\ q_3 & -q_4 \\ -p_3 & p_4 \end{bmatrix}, \dots$$

Then the p_k, q_k found in this way are the same as found in the convergents (13).

Proof. Let $\tilde{p}_k, \tilde{q}_k$ denote the p_k, q_k obtained in (27). We show that $\tilde{p}_k = p_k$ and $\tilde{q}_k = q_k$ by induction on k , the cases $k = 1, 2$ being trivial.

First let k be even. By (25), $\tilde{p}_{k+1} = \tilde{p}_{k-1} + t\tilde{p}_k$ and $\tilde{q}_{k+1} = \tilde{q}_{k-1} + t\tilde{q}_k$, where $t = \lfloor \alpha_k / \beta_k \rfloor$. By (26), $\alpha_k = \tilde{q}_{k-1}\alpha - \tilde{p}_{k-1}$ and $\beta_k = -\tilde{q}_k\alpha + \tilde{p}_k$. Hence t is the largest integer such that

$$(28) \quad t \leq \frac{\tilde{q}_{k-1}\alpha - \tilde{p}_{k-1}}{-\tilde{q}_k\alpha + \tilde{p}_k}, \quad \text{i.e.} \quad \frac{\tilde{p}_{k-1} + t\tilde{p}_k}{\tilde{q}_{k-1} + t\tilde{q}_k} \leq \alpha.$$

So the induction hypothesis: $\tilde{p}_{k-1} = p_{k-1}$, $\tilde{q}_{k-1} = q_{k-1}$, $\tilde{p}_k = p_k$, $\tilde{q}_k = q_k$, implies $\tilde{p}_{k+1} = p_{k+1}$ and $\tilde{q}_{k+1} = q_{k+1}$.

The case of k odd is shown similarly. $\square$

So from the Euclidean algorithm and Theorem 5.1 we know that if α is rational, the continued fraction method terminates, in polynomial time. Therefore, we have the following corollary.

Corollary 6.2a. *Given rationals α and ε ($0 < \varepsilon \leq 1$), the continued fraction method finds integers p and q as described in Dirichlet's theorem, in time polynomially bounded by the size of α .*

Proof. Directly from the above and Theorem 6.2 (if $\alpha < 0$, applied to $-\alpha$). $\square$

The following variant of Theorem 6.2 is due to Legendre [1798: pp. 27–29].

Theorem 6.3. *Let $\alpha \geq 0$ be a real number, and let p and q be natural numbers with $|\alpha - p/q| < 1/2q^2$. Then p/q occurs as convergent for α .*

Proof. The proof uses the fact that if $p'/q' < p''/q''$ are rational numbers ($p', p'' \in \mathbb{Z}$, $q', q'' \in \mathbb{N}$), then

$$(29) \quad \frac{p''}{q''} - \frac{p'}{q'} = \frac{p''q' - p'q''}{q'q''} \geq \frac{1}{q'q''}.$$

To prove the theorem, first suppose that $p/q < \alpha$. Make the continued fraction expansion $p_1/q_1, p_2/q_2, \dots$ of α , and let p_i/q_i be that convergent of α with $p_i/q_i \leq p/q$ and i as large as possible (so i is odd). We show that $p_i/q_i = p/q$. Suppose to the contrary that $p_i/q_i < p/q$. Then there exists an integer $u \geq 0$ such that

$$(30) \quad \frac{p_i + up_{i+1}}{q_i + uq_{i+1}} \leq \frac{p}{q} < \frac{p_i + (u+1)p_{i+1}}{q_i + (u+1)q_{i+1}} \leq \alpha.$$

This is easy to see if $p_{i+1}/q_{i+1} = \alpha$. If $p_{i+1}/q_{i+1} > \alpha$, then by definition of p_{i+2}/q_{i+2} there exists an integer $t \geq 1$ with

$$(31) \quad \alpha \geq \frac{p_i + tp_{i+1}}{q_i + tq_{i+1}} = \frac{p_{i+2}}{q_{i+2}} > \frac{p}{q}$$

which also implies the existence of an integer $u \geq 0$ satisfying (30).

Now $q_i + uq_{i+1} \leq q$, as

$$(32) \quad \frac{1}{qq_{i+1}} \leq \frac{p_{i+1}}{q_{i+1}} - \frac{p}{q} \leq \frac{p_{i+1}}{q_{i+1}} - \frac{p_i + up_{i+1}}{q_i + uq_{i+1}} = \frac{1}{q_{i+1}(q_i + uq_{i+1})}$$

(by (29), (30), and since $p_{i+1}q_i - p_iq_{i+1} = 1$). Also $q_{i+1} \leq q$, as

$$(33) \quad \frac{1}{qq_i} \leq \frac{p}{q} - \frac{p_i}{q_i} \leq \frac{p_{i+1}}{q_{i+1}} - \frac{p_i}{q_i} = \frac{1}{q_{i+1}q_i}$$

(by (29) and since $p_{i+1}q_i - p_iq_{i+1} = 1$).

Therefore, $q_i + (u+1)q_{i+1} \leq 2q$. Hence

$$(34) \quad \alpha - \frac{p}{q} \geq \frac{p_i + (u+1)p_{i+1}}{q_i + (u+1)q_{i+1}} - \frac{p}{q} \geq \frac{1}{q(q_i + (u+1)q_{i+1})} \geq \frac{1}{2q^2},$$

which contradicts our assumption.

The case $p/q > \alpha$ is shown similarly. □

Theorem 6.3 implies that if a rational number is near enough to a rational number of low denominator, this last rational can be found in polynomial time. More precisely:

Corollary 6.3a. *There exists a polynomial algorithm which, for given rational number α and natural number M , tests if there exists a rational number p/q with $1 \leq q \leq M$ and $|\alpha - p/q| < 1/2M^2$, and if so, finds this (unique) rational number.*

Proof. It is clear that if such a rational number p/q exists, it is unique, for if p'/q' is another such rational number, we have the contradiction:

$$(35) \quad \frac{1}{M^2} \leq \frac{1}{qq'} \leq \left| \frac{p}{q} - \frac{p'}{q'} \right| \leq \left| \alpha - \frac{p}{q} \right| + \left| \alpha - \frac{p'}{q'} \right| < \frac{1}{2M^2} + \frac{1}{2M^2} = \frac{1}{M^2}.$$

First let $\alpha > 0$. Suppose p/q as required exists. As

$$(36) \quad \left| \alpha - \frac{p}{q} \right| < \frac{1}{2M^2} \leq \frac{1}{2q^2}$$

Theorem 6.3 implies that p/q occurs as convergent of α . So it suffices to compute the convergents of α (as long as $q_i \leq M$), and to test if one of them satisfies (36).

If $\alpha < 0$ we apply the above to $-\alpha$, while for $\alpha = 0$ the algorithm is trivial. □

The continued fraction method approximates one number. In Section 6.3 we shall see an extension to the approximation of vectors, i.e. to approximating several real numbers simultaneously, by rationals with one common low denominator: *simultaneous diophantine approximation*.

Notes for Section 6.1. Papadimitriou [1979] and Reiss [1979] (cf. Zemel [1981–2]) derived with continued fractions and the related Farey series that there exists an algorithm

which identifies a number $s^* = p/q$, with $p, q \in \mathbb{N}$ and $p, q \leq M$, within $O(\log M)$ queries of the form: 'is $s^* \leq s$?' Indeed, it is clear that, by binary search, within $\lceil \log_2(2M^3) \rceil$ queries we can find $m \in \mathbb{N}$ such that $1 \leq m \leq 2M^3$ and

$$(37) \quad \frac{m-1}{2M^2} < s^* \leq \frac{m}{2M^2}$$

(as $0 < s^* \leq M$). Now apply Corollary 6.3a to $x := m/2M^2$.

Ursic and Patarra [1983] apply continued fractions to finding an exact solution for linear equations if an approximate solution is available.

An elementary exposition of continued fractions is given by Richards [1981].

The average length of the continued fraction method is studied by Heilbronn [1968], Dixon [1970], and Tonkov [1974]. For more on continued fractions, see Cheng and Pollington [1981] and Nathanson [1974].

Hurwitz [1891] showed that for each irrational number x , there exist infinitely many convergents p/q in the continued fraction approximation for x such that

$$(38) \quad \left| x - \frac{p}{q} \right| < \frac{1}{\sqrt{5} q^2}$$

and that the factor $1/\sqrt{5}$ here is best possible (as is shown by $x = \frac{1}{2} + \frac{1}{2}\sqrt{5}$). See Cohn [1973] for a short proof.

6.2. BASIS REDUCTION IN LATTICES

Let A be a nonsingular matrix of order n , and let Λ be the lattice generated by the columns of A . From Chapter 4 we know that if B is another nonsingular matrix, whose columns also generate Λ , then $|\det A| = |\det B|$. So this number is independent of the choice of the basis, and is called the *determinant* of Λ , denoted by $\det \Lambda$. It is equal to the volume of the parallelepiped $\{\lambda_1 b_1 + \dots + \lambda_n b_n \mid 0 \leq \lambda_i < 1 \text{ for } i = 1, \dots, n\}$, where $b_1, \dots, b_n$ is any basis for Λ . This gives the well-known *Hadamard inequality*:

$$(39) \quad \det \Lambda \leq \|b_1\| \cdots \|b_n\|$$

where $\|\cdot\|$ denotes Euclidean norm ($\|x\| = \sqrt{x^T x}$).

It is trivial that equality in (39) occurs only if $b_1, \dots, b_n$ are orthogonal, and that not every lattice has an orthogonal basis. A classical theorem of Hermite [1850a] states that for each n there exists a number $c(n)$ such that every n -dimensional lattice Λ has a basis $b_1, \dots, b_n$ with

$$(40) \quad \|b_1\| \cdots \|b_n\| \leq c(n) \det \Lambda.$$

Such a basis may be viewed as an approximation of an orthogonal basis.

Hermite showed that we can take

$$(41) \quad c(n) = \left(\frac{4}{3}\right)^{n(n-1)/4}.$$

Minkowski [1896] improved this by showing that even

$$(42) \quad c(n) = 2^n / V_n \approx \left(\frac{2n}{\pi e}\right)^{n/2}$$

works, where V_n denotes the volume of the n -dimensional unit ball. However, for choice (42) of $c(n)$ no polynomial algorithm finding a basis satisfying (40) is known. But if we take $c(n) = 2^{n(n-1)/4}$, Lovász designed a polynomial algorithm finding such a basis (if the lattice is given by generators). In fact, Schnorr [1985] proved that for each fixed $\varepsilon > 0$ there exists a polynomial algorithm finding a basis satisfying (40) with $c(n) = (1 + \varepsilon)^{n(n-1)}$, if $n \geq n(\varepsilon)$.

We now first describe Lovász's basis reduction method (see Lenstra, Lenstra, and Lovász [1982]) in terms of positive definite matrices (the above formulation comes back in Corollary 6.4a).

Theorem 6.4 (Basis reduction method). *There exists a polynomial algorithm which, for given positive definite rational matrix D , finds a basis $b_1, \dots, b_n$ (where $n := \text{order}(D)$) for the lattice $\mathbb{Z}^n$ satisfying*

$$(43) \quad \|b_1\| \cdots \|b_n\| \leq 2^{n(n-1)/4} \sqrt{\det D},$$

where $\|x\| := \sqrt{x^T D x}$.

Proof. I. We describe the algorithm. Let the positive definite $n \times n$ -matrix D be given. Denote by $\perp$ orthogonality relative to the inner product defined by D . So:

$$(44) \quad x \perp y \text{ if and only if } x^T D y = 0.$$

Without loss of generality we assume that D is integral (otherwise, multiply D by an appropriate integer).

Construct a series of bases for $\mathbb{Z}^n$ as follows. The unit basis vectors form the first basis. Suppose inductively that we have found a basis $b_1, \dots, b_n$ for $\mathbb{Z}^n$. Now apply the following Steps 1, 2, and 3.

Step 1. Denote by B_i the matrix with columns $b_1, \dots, b_i$. Calculate

$$(45) \quad b_i^* = b_i - B_{i-1}(B_{i-1}^T D B_{i-1})^{-1} B_{i-1}^T D b_i.$$

Comment 1. The vectors $b_1^*, \dots, b_n^*$ form the *Gram Schmidt orthogonalization* of $b_1, \dots, b_n$ relative to $\perp$. That is, for $i = 1, \dots, n$, b_i^* is the unique vector satisfying

$$(46) \quad b_i^* \perp b_1, \dots, b_{i-1}, \text{ and } b_i^* - b_i \in \text{lin.hull}\{b_1, \dots, b_{i-1}\}.$$

Geometrically, b_i^* is the perpendicular line (relative to $\perp$) from b_i to $\text{lin.hull}\{b_1, \dots, b_{i-1}\} = \text{lin.hull}\{b_1^*, \dots, b_{i-1}^*\}$. The vectors $b_1^*, \dots, b_n^*$ are pairwise orthogonal (relative to $\perp$), and for each i ,

$$(47) \quad b_i = \lambda_{i1} b_1^* + \cdots + \lambda_{i,i-1} b_{i-1}^* + b_i^*$$

for some $\lambda_{i1}, \dots, \lambda_{i,i-1}$.

Step 2. For $i = 2, \dots, n$, do the following: for $j = i-1, i-2, \dots, 1$, do: write $b_i = \lambda_{i1} b_1^* + \cdots + \lambda_{i,i-1} b_{i-1}^* + b_i^*$, and replace b_i by $b_i - \lfloor \lambda_{ij} + \frac{1}{2} \rfloor b_j$.

Comment 2. In matrix language, Step 2 does the following. Let C_i be the matrix with

columns $b_1^*, \dots, b_n^*$. Then $B_n = C_n V$ for some upper-triangular matrix V , with 1's on the main diagonal (by (47)). By elementary column operations one can change V into a matrix W in upper-triangular form, with 1's on the diagonal, and all other entries at most $\frac{1}{2}$ in absolute value. Then the columns of $C_n W$ form the basis $b_1, \dots, b_n$ as it is after applying Step 2.

Step 3. Choose, if possible, an index i such that $\|b_i^*\|^2 > 2\|b_{i+1}^*\|^2$. Exchange b_i and b_{i+1} , and start with Step 1 anew.

If no such i exists, the algorithm stops.

Two further comments:

Comment 3. The replacements made in Step 2 do not change the Gram-Schmidt orthogonalization of $b_1, \dots, b_n$. Note that at completion of Step 2, we have for $i = 1, \dots, n$,

$$(48) \quad b_i = \lambda_1 b_1^* + \dots + \lambda_{i-1} b_{i-1}^* + b_i^*$$

with $|\lambda_j| \leq \frac{1}{2}$ for $j = 1, \dots, i-1$, since $B_n = C_n W$.

Comment 4. In the notation of Step 1,

$$(49) \quad \|b_1^*\|^2 \cdots \|b_n^*\|^2 = \det(B_1^T D B_1).$$

Indeed, let C_i be the matrix with columns $b_1^*, \dots, b_i^*$. As in Comment 2, there is an upper triangular matrix V_i with 1's on the diagonal such that $C_i V_i = B_i$. Then, as $b_i^* \perp b_j^*$ if $i \neq j$,

$$(50) \quad \|b_1^*\|^2 \cdots \|b_i^*\|^2 = \det(C_i^T D C_i) = \det(V_i^T C_i^T D C_i V_i) = \det(B_i^T D B_i).$$

In particular, as $|\det B_n| = 1$,

$$(51) \quad \|b_1^*\|^2 \cdots \|b_n^*\|^2 = \det D.$$

II. If the algorithm stops (i.e. the choice of i in Step 3 is not possible), we have $\|b_i^*\|^2 \leq 2\|b_{i+1}^*\|^2$ for $i = 1, \dots, n-1$. Hence, inductively, $\|b_i^*\|^2 \leq 2^{k-1}\|b_k^*\|^2$ for $1 \leq i < k \leq n$. By Comments 1 and 3,

$$(52) \quad \|b_k\|^2 \leq \|b_k^*\|^2 + \frac{1}{4} \sum_{i=1}^{k-1} \|b_i^*\|^2 \leq \|b_k^*\|^2 \left(1 + \frac{1}{4} \sum_{i=1}^{k-1} 2^{k-i} \right) \leq 2^{k-1} \|b_k^*\|^2.$$

Therefore, using (51),

$$(53) \quad \|b_1\|^2 \cdots \|b_n\|^2 \leq \prod_{k=1}^n (2^{k-1} \|b_k^*\|^2) = 2^{n(n-1)/2} \det D.$$

So the final basis $b_1, \dots, b_n$ satisfies (43).

III. We still have to show that the algorithm terminates, in polynomial time. Consider the following function (using the notation of Step 1):

$$(54) \quad F(b_1, \dots, b_n) := \det(B_1^T D B_1) \det(B_2^T D B_2) \cdots \det(B_n^T D B_n).$$

Then, by Comment 4,

$$(55) \quad F(b_1, \dots, b_n) = \|b_1^*\|^{2n} \cdot \|b_2^*\|^{2n-2} \cdot \|b_3^*\|^{2n-4} \cdots \|b_n^*\|^2.$$

So at Step 2, $F(b_1, \dots, b_n)$ does not change (cf. Comment 3).

If at Step 3 we interchange b_i and b_{i+1} , then only the term $\det(B_i^T DB_i)$ of (54) changes: the matrices $B_1, \dots, B_{i-1}$ do not change at all, while in the matrices $B_{i+1}, \dots, B_n$ two columns are interchanged, which leaves the corresponding terms of (54) invariant.

To avoid confusion, let $\tilde{b}_1, \dots, \tilde{b}_n$ be the new basis (after application of Step 3), and let $\tilde{b}_1^*, \dots, \tilde{b}_n^*$ be the new Gram-Schmidt orthogonalization. Then $\tilde{b}_j = b_j$ for $j \neq i, i+1$, and $\tilde{b}_i = b_{i+1}$ and $\tilde{b}_{i+1} = b_i$. So $\tilde{b}_j^* = b_j^*$ for $j \neq i, i+1$. Therefore,

$$(56) \quad \begin{aligned} \tilde{b}_i &= b_{i+1} = \lambda_1 b_1^* + \dots + \lambda_{i-1} b_{i-1}^* + \lambda_i b_i^* + b_{i+1}^* \\ &= \lambda_1 \tilde{b}_1^* + \dots + \lambda_{i-1} \tilde{b}_{i-1}^* + \lambda_i b_i^* + b_{i+1}^* \end{aligned}$$

with the λ_j at most $\frac{1}{2}$ in absolute value. As $\lambda_i b_i^* + b_{i+1}^* \perp b_1, \dots, b_{i-1}$, we know that $\tilde{b}_i^* = \lambda_i b_i^* + b_{i+1}^*$. Therefore, by our choice of i in Step 3:

$$(57) \quad \frac{\|\tilde{b}_i^*\|^2}{\|b_i^*\|^2} = \frac{\lambda_i^2 \|b_i^*\|^2 + \|b_{i+1}^*\|^2}{\|b_i^*\|^2} = \lambda_i^2 + \frac{\|b_{i+1}^*\|^2}{\|b_i^*\|^2} < \frac{1}{4} + \frac{1}{2} = \frac{3}{4}.$$

So by interchanging b_i and b_{i+1} , $\det(B_i^T DB_i)$ is multiplied by a factor $\mu < 3/4$. Hence, at Step 3, the value of $F(b_1, \dots, b_n)$ is multiplied by $\mu < 3/4$.

Let T be the maximum absolute value of the entries of D . Then initially, and hence by the above throughout the procedure, we have:

$$(58) \quad \det(B_i^T DB_i) \leq (nT)^n$$

and hence

$$(59) \quad F(b_1, \dots, b_n) \leq (nT)^{n^2}.$$

Since $F(b_1, \dots, b_n)$ is a positive integer (by (54), as $b_1, \dots, b_n$ are linearly independent integral vectors, and D is integral), the algorithm terminates after at most

$$(60) \quad \log_{4/3}((nT)^{n^2}) = n^2(\log_2 n + \log_2 T) \log_{4/3} 2.$$

iterations, which number is polynomially bounded by the size of D .

IV. We still must show that the sizes of the intermediate numbers also are polynomially bounded by the size of D . From (45) it follows that throughout the algorithm, the vectors $\det(B_{i-1}^T DB_{i-1}) \cdot b_i^*$ are integral. Hence the denominators in b_i^* are at most $\det(B_{i-1}^T DB_{i-1}) \leq (nT)^n$ (cf. (58)). This implies

$$(61) \quad \|b_i^*\| \geq (nT)^{-n}$$

where $\|\cdot\|$ denotes the Euclidean norm ($\|x\| = \sqrt{x^T x}$). Let λ denote the smallest eigenvalue of D . Then, again denoting the maximum absolute value of the entries of D by T ,

$$(62) \quad \lambda^{-1} = \text{largest eigenvalue of } D^{-1} \leq \text{Tr}(D^{-1}) \leq n(nT)^n$$

(as the entries of D^{-1} are quotients of (sub)determinants of D , which are at most $(nT)^n$). Therefore, by (49), (58), and (61),

$$(63) \quad \begin{aligned} \|b_i^*\|^2 &\leq \lambda^{-1} \|b_i^*\|^2 = \lambda^{-1} \det(B_i^T DB_i) \|b_1^*\|^{-2} \dots \|b_{i-1}^*\|^{-2} \\ &\leq \lambda^{-n} \det(B_i^T DB_i) \|b_1^*\|^{-2} \dots \|b_{i-1}^*\|^{-2} \\ &\leq n^n (nT)^{n^2} (nT)^{2n^2} \leq (nT)^{3n^2 + 2n}. \end{aligned}$$

Hence the sizes of the numerators and denominators in the b_i^* are polynomially bounded by the size of D .

After application of Step 2 we know by Comment 3:

$$(64) \quad \|b_i\| \leq \|b_1^*\| + \cdots + \|b_i^*\| \leq n(nT)^{3n^2+2n}.$$

Since the b_i are integral vectors, it follows that also the sizes of the b_i are polynomially bounded by the size of D . $\square$

Corollary 6.4a. *There exists a polynomial algorithm which, for given nonsingular rational matrix A , finds a basis $b_1, \dots, b_n$ (with $n := \text{order}(A)$) for the lattice Λ generated by the columns of A satisfying*

$$(65) \quad \|b_1\| \cdots \|b_n\| \leq 2^{n(n-1)/4} |\det A|$$

where $\|x\| = \sqrt{x^T x}$.

Proof. Apply the basis reduction algorithm (Theorem 6.4) to the matrix $D := A^T A$. This gives a basis $b_1, \dots, b_n$ for $\mathbb{Z}^n$ with

$$(66) \quad \|Ab_1\| \cdots \|Ab_n\| = \sqrt{b_1^T D b_1} \cdots \sqrt{b_n^T D b_n} \leq 2^{n(n-1)/4} \sqrt{\det D} \\ = 2^{n(n-1)/4} |\det A|.$$

The vectors $Ab_1, \dots, Ab_n$ form a basis for Λ , and the corollary follows. $\square$

Notes for Section 6.2. In fact, Hermite's factor $c(n) = (\frac{4}{3})^{n(n-1)/4}$ in (40) can be approached arbitrarily close by a slight adaptation of the above algorithm. Then one shows: *there exists an algorithm which, for given positive definite rational matrix D and given $\epsilon > 0$, finds a basis $b_1, \dots, b_n$ for the lattice $\mathbb{Z}^n$ satisfying*

$$(67) \quad \|b_1\| \cdots \|b_n\| \leq (\frac{4}{3} + \epsilon)^{n(n-1)/4} \sqrt{\det D}$$

in time polynomially bounded by size(D) and by $1/\epsilon$.

Kannan [1983b] gave an improved version of the basis reduction method, from which Schnorr [1985] derived that for each fixed $\epsilon > 0$ one can find a basis $b_1, \dots, b_n$ for $\mathbb{Z}^n$ with

$$(68) \quad \|b_1\| \cdots \|b_n\| \leq (1 + \epsilon)^{n(n-1)/4} \sqrt{\det D}$$

in time polynomially bounded by size(D).

Lenstra [1983] showed with methods from the geometry of numbers that there exists a constant c such that for each fixed n , there is a polynomial-time algorithm finding, for given positive definite rational $n \times n$ -matrix D , a basis $b_1, \dots, b_n$ for $\mathbb{Z}^n$ such that

$$(69) \quad \|b_1\| \cdots \|b_n\| \leq (cn)^n \sqrt{\det D}.$$

The order of this bound is close to that of Minkowski's bound (42).

6.3 APPLICATIONS OF THE BASIS REDUCTION METHOD

Finding a short nonzero vector in a lattice

It is a classical result of Minkowski [1891] that any n -dimensional lattice Λ contains a nonzero vector b with

$$(70) \quad \|b\| \leq 2 \left(\frac{\det \Lambda}{V_n} \right)^{1/n},$$

where again V_n denotes the volume of the n -dimensional unit ball.

Sketch of proof. Let $r := \frac{1}{2} \min \{ \|b\| \mid b \in \Lambda; b \neq 0 \}$. Then for $x, y \in \Lambda$ with $x \neq y$ one has $B^0(x, r) \cap B^0(y, r) = \emptyset$ (where $B^0(x, r)$ denotes the open ball of radius r with centre x). Now $B^0(x, r)$ has volume $r^n V_n$. Hence the density of $\cup_{x \in \Lambda} B^0(x, r)$ is $r^n V_n (\det \Lambda)^{-1}$, which is at most 1. This gives (70). $\square$

To date, no polynomial algorithm finding a vector b satisfying (70) is known. It is conjectured that finding a shortest nonzero vector in Λ (given by a basis) is $\mathcal{NP}$ -complete.

However, with the basis reduction method of Theorem 6.4 one can find a 'longer short vector' in a lattice, by taking the shortest vector in the basis constructed there.

Corollary 6.4b. *There exists a polynomial algorithm which, given a nonsingular rational matrix A , finds a nonzero vector b in the lattice Λ generated by the columns of A with*

$$(71) \quad \|b\| \leq 2^{n(n-1)/4} \cdot (\det \Lambda)^{1/n}$$

where $n := \text{order}(A)$.

Proof. The shortest vector in the basis $b_1, \dots, b_n$ described in Corollary 6.4a has length at most

$$(72) \quad (2^{n(n-1)/4} \det \Lambda)^{1/n} = 2^{(n-1)/4} (\det \Lambda)^{1/n}.$$

[In fact, if we follow the algorithm as described in the proof of Theorem 6.4, and the notation of the proof of Corollary 6.4a, we can take Ab_1 for the vector b satisfying (71). Indeed, in part II of the proof of Theorem 6.4 we saw that $\|b_i^*\|^2 \leq 2^{k-i} \|b_k^*\|^2$ for $1 \leq i < k \leq n$. Moreover, by Comment 4 in that proof, $\|b_1^*\| \dots \|b_n^*\| = \det \Lambda$. Hence,

$$(73) \quad \|Ab_1\| = \|b_1\| = \left(\prod_{k=1}^n \|b_k^*\| \right)^{1/n} \leq \left(\prod_{k=1}^n 2^{(k-1)/2} \|b_k^*\| \right)^{1/n} \\ = (2^{n(n-1)/4} \det \Lambda)^{1/n} \quad \square$$

So with the basis reduction method also a short nonzero vector can be found, albeit generally not the shortest.

Van Emde Boas [1981] proved that, for any fixed norm, the problem: 'given lattice Λ and vector a , find $b \in \Lambda$ with norm of $b - a$ as small as possible' is $\mathcal{NP}$ -complete (it is the integer analogue of the, polynomially solvable, least squares problem).

Algorithms for finding shortest vectors in lattices were described by Rosser [1942], Coveyou and MacPherson [1967], and Knuth [1969: pp. 89–98] (cf. also Dieter [1975]). Kannan [1983b] gave an algorithm for finding a shortest nonzero lattice vector using a variant of Lovász's basis reduction method.

Simultaneous diophantine approximation

Another application is to *simultaneous diophantine approximation*. Extending the 'pigeon-hole' argument of Theorem 6.1, Dirichlet [1842] showed that if

$$(74) \quad \left| x_i - \frac{p_i}{q} \right| < \frac{\varepsilon}{q}, \quad \text{for } i = 1, \dots, n, \quad \text{and } 1 \leq q \leq \varepsilon^{-n}.$$

However, a consequence of Corollary 6.4b is that a weaker approximation can be found in polynomial time, which turns out to be strong enough, for example, for application in the ellipsoid method in Chapter 14.

$$(75) \quad \left\| a - \frac{1}{q} p \right\| < \frac{\varepsilon}{q}, \quad \text{and} \quad 1 \leq q \leq 2^{n(n+1)/4} \varepsilon^{-n}.$$
$$(76) \quad A := \left[\begin{array}{c|c} \begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array} & \begin{array}{c} -\alpha_1 \\ \vdots \\ -\alpha_n \end{array} \\ \hline \begin{array}{cc} 0 & 0 \end{array} & 2^{-n(n+1)/4} \varepsilon^{n+1} \end{array} \right]$$
$$(77) \quad \|h\| \leq 2^{n+4} (2^{-n(n+1)+4} \varepsilon^{n+1})^{1/(n+1)} = \varepsilon.$$
$$(78) \quad \begin{bmatrix} p_1 \\ \vdots \\ p_n \\ q \end{bmatrix} := A^{-1}b$$
$$(79) \quad b = A \begin{bmatrix} p_1 \\ \vdots \\ p_n \\ q \end{bmatrix}.$$

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Without loss of generality, $q \geq 1$ (otherwise, replace b by $-b$). Then

$$(80) \quad \beta_i = p_i - q\alpha_i, \quad \text{for } i = 1, \dots, n, \quad \text{and} \quad \beta_{n+1} = q \cdot 2^{-(n+1)/4} \varepsilon^{n+1}.$$

Hence, defining $p := (p_1, \dots, p_n)^T$,

$$(81) \quad \|q \cdot a - p\| < \|b\| \leq \varepsilon \quad \text{and} \quad q = \beta_{n+1} \cdot 2^{n(n+1)/4} \varepsilon^{-n-1} \leq 2^{n(n+1)/4} \varepsilon^{-n}$$

(using $\beta_{n+1} \leq \|b\| \leq \varepsilon$). □

Lagarias [1982] showed that the problem: 'Given rationals $\alpha_1, \dots, \alpha_n$, a natural number M , and a rational $\varepsilon > 0$, decide if there exist integers $p_1, \dots, p_n, q$ such that $|x_i - (p_i/q)| \leq \varepsilon/q$ and $1 \leq q \leq M$ ' is $\mathcal{NP}$ -complete. He showed moreover that, if we fix n , this problem is polynomially solvable. Lagarias also gave some sharpenings of Corollary 6.4c.

Finding the Hermite normal form

Another application of the basis reduction method is to finding the Hermite normal form of a matrix, in polynomial time. Let A be a nonsingular integral matrix of order n . Define

$$(82) \quad M := \lceil 2^{n(n-1)/4} |\det A| \rceil$$

and let C arise from A by multiplying the i th row of A by M^{n-i} , for $i = 1, \dots, n$. By Corollary 6.4a one can find in polynomial time a basis $b_1, \dots, b_n$ for the lattice generated by the columns of C satisfying

$$(83) \quad \|b_1\| \cdots \|b_n\| \leq 2^{n(n-1)/4} |\det C| = 2^{n(n-1)/4} M^{n(n-1)/2} |\det A|.$$

Then the vectors $b_1, \dots, b_n$ can be reordered in such a way that the matrix $[b_1, \dots, b_n]$ is in lower triangular form. Indeed, by reordering we may assume that the j th coordinate of b_j is nonzero, for $j = 1, \dots, n$. So the j th coordinate of b_j is at least M^{n-j} in absolute value, and hence $\|b_j\| \geq M^{n-j}$. Suppose that the i th coordinate of b_j is nonzero, for some $1 \leq i < j \leq n$. Then $\|b_j\| > M^{n-i} \geq M \cdot M^{n-j}$, and

$$(84) \quad \|b_1\| \cdots \|b_n\| > \left(\prod_{j=1}^n M^{n-j} \right) \cdot M \geq 2^{n(n-1)/4} M^{n(n-1)/2} |\det A|$$

contradicting (83).

So $b_1, \dots, b_n$ can be reordered so that the matrix $[b_1, \dots, b_n]$ is in lower triangular form. This matrix can be easily modified by elementary column operations to Hermite normal form. Dividing the j th row by M^{n-j} (for each j) gives the Hermite normal form of A .

Other applications of the basis reduction method appear in Lenstra's integer linear programming algorithm (Section 18.4), in factoring polynomials (over the rationals) in polynomial time (see Lenstra, Lenstra, and Lovász [1982]), in breaking cryptographic codes (see Shamir [1982], Adleman [1983a, b], and Lagarias and Odlyzko [1983]), in disproving *Mertens' conjecture* (Odlyzko and te Riele [1984]), and in solving low density subset sum problems (Lagarias and Odlyzko [1983]).

Notes for Chapter 6. For more about more-dimensional continued fractions and simultaneous diophantine approximation, see Adams [1967, 1969a, b, 1971], Bernstein [1971], Brentjes [1981], Brun [1959, 1961], Bundschuh [1975], Cassels [1955], Cusick [1971, 1972a, b, 1974, 1977a, b, 1980], Davenport [1955], Davenport and Mahler [1946], Davenport and Schmidt [1967, 1969, 70, 1970], Dubois and Rhin [1980, 1982], Ferguson and Forcade [1979], Fischer and Schweiger [1975], Furtwängler [1927, 1928], Güting [1975, 1976a, b], Jacobi [1868], Jurkat, Kratz and Peyerimhoff [1977, 1979], Jurkat and

Peyerimhoff [1976], Koksma and Meulenbeld [1941], Lagarias [1980a], Langmayr [1979a, b], Mack [1977], Mahler [1976], Mullender [1950], Nowak [1981], Paley and Ursell [1930], Peck [1961], Perron [1907], Pipping [1921, 1957], Podsypanin [1977], Poincaré [1884], Raisbeck [1950], Schmidt [1966], Schweiger [1973, 1977], Selmer [1961], Spohn [1968], Szekeres [1970], Thurnheer [1981], Tichy [1979], Vaughan [1978], and Wills [1968a, b, 1970].

For a survey on basis reduction algorithms, see Bachem and Kannan [1984].

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Notes on lattices and linear diophantine equations

HISTORICAL NOTES

The Ancient Civilizations

The number theory discussed in this Part is to a large extent quite classic.

The ancient Greeks knew the Euclidean algorithm and the continued fraction method. The Euclidean algorithm was described in Book VII of Euclid's *Elements* (written $\pm$ 330–320 B.C.). Books V and X of the *Elements* describe the continued fraction method to test if two given lengths have a 'common measure'. Two ratios are defined to be equal if they have the same continued fraction expansion. The method was called ἀνθυφαίρεῖν ('reciprocal elimination') or ἀντὶἀνυφαίρεῖν ('cancellation of items on an account'). In the method, the larger number was not divided by the smaller, but the smaller was subtracted from the larger repeatedly.

These Books of the *Elements* are based on earlier work of Theaetetus (c. 415–c. 369 B.C.) (cf. Becker [1933]). Also Eudoxus (c. 408–c. 355 B.C.) and Aristotle (384–322 B.C.) knew the continued fraction method as a way of testing equality of two ratios. The method was used by Aristarchus of Samos (c. 310–c. 230 B.C.) and Archimedes (287–212 B.C.), the latter for determining square roots.

The Euclidean algorithm was also described in Book I of the *Chiu-Chang Suan-Shu* ('Nine Books on Arithmetic'). Book II of this collection discusses some special classes of linear diophantine equations. The *Sun-Tzu Suan-Ching* ('Master Sun's Arithmetic Manual'), written in China between A.D. 280 and 473, considers the 'Chinese remainder problem', which consists of a special type of system of linear diophantine equations (cf. Midonick [1968: pp. 211–218]).

Diophantos of Alexandria ($\pm$ 3rd century) has yielded the name *diophantine equation*. In his book *Arithmetika* he treats the problem of finding nonnegative rational solutions to equations (mainly quadratic), where, by making the equations homogeneous, the problem can be reduced to finding nonnegative integral solutions. He seems not to have considered solving linear equations in integers.

In fact, the solution of linear diophantine equations has not been found in any ancient Greek text. But, according to van der Waerden [1983], excellent Greek mathematicians like Archimedes or Apollonius would have had no difficulty in solving linear diophantine equations by means of the Euclidean algorithm.

Between 200 and 1200 several Indian mathematicians gave methods for solving linear diophantine equations in two unknowns. The earliest known systematic treatment of linear diophantine equations in two variables was given by Āryabhata (476–550), who proposed a continued-fraction-like method to find all integral solutions of $ax + by = c$. Āryabhata was an astronomer, interested in linear diophantine equations to determine periods of astronomical conjunctions. He and his followers, like Brahmagupta (b. 598), Bhāscara (629), Bhāscara Āchārya (1114–c. 1185) (cf. Midonick [1968: pp. 270–274]), and Parameshvara (1431), also considered special types of systems of two linear diophantine equations (cf. van der Waerden [1983: pp. 112–120], Kline [1972: pp. 184–188], Dickson [1920: pp. 41–44]).

16th to 18th century in Europe

In Europe, in the 17th and 18th century, solving linear diophantine equations in two or three variables, with reduction methods like the Euclidean algorithm, was studied, *inter alia*, by Bachet [1612], Kersey [1673], Rolle [1690], De Lagny [1697, 1733], Euler [1740; 1770a: 2.Theil, 2.Abschnitt, Capitel I], Saunderson [1741], and Lagrange [1769a: Art. 7–8; 1769b; 1770: Lemme 1, §24; 1774: Section III; 1798] (cf. Dickson [1920: pp. 44–99]).

After some isolated work on *continued fractions* in the 16th and 17th century (Bombelli, Brouncker, Cataldi, Huygens), a systematic theory of continued fractions, as a main tool for approximation, was developed in the 18th century by Euler [1744], Lagrange [1769b, 1774, 1798], Lambert, and Legendre [1798]. Lagrange [1774: Section II] showed:

- (1) for any convergent p/q in the continued fraction method for a real number x one has $|p - xq| < |p' - xq'|$ whenever $p' < p$ and $q' < q$

(cf. Smith [1925: pp. 418–421], Jones and Thron [1980: pp. 1–11]).

Lagrange on binary quadratic forms

Lagrange [1775a] started the systematic study of *quadratic forms* in integer variables, following Fermat's work on special quadratic forms. Important is the concept of equivalence: two quadratic forms $x^T D_1 x$ and $x^T D_2 x$ (or two matrices D_1 and D_2) are *equivalent* if there exists a unimodular matrix U with $U^T D_1 U = D_2$. So if D_1 and D_2 are equivalent, $x^T D_1 x = t$ is solvable in integers if and only if $x^T D_2 x = t$ is also.

Lagrange [1775a] studied the problem of *reducing* a binary quadratic form (*binary* means: in two variables). He showed that each positive definite 2×2 -matrix D is equivalent to a matrix $\begin{bmatrix} a & b \\ b & c \end{bmatrix}$ with $|2b| \leq a \leq c$. A matrix with this last property was called by Lagrange *reduced*. It implies

- (2) $ac \leq \frac{4}{3} \det D$

or, equivalently, that each lattice Λ in two dimensions has a basis b_1, b_2 with $\|b_1\| \cdot \|b_2\| \leq \sqrt{\frac{4}{3} \det \Lambda}$.

So each quadratic form $ax^2 + 2bxy + cy^2$ with $a > 0$, $ac - b^2 > 0$ is equivalent to a quadratic form $a'x^2 + 2b'xy + c'y^2$ with $|2b'| \leq a' \leq c'$. As for given $\Delta > 0$ there are only finitely many integral triples a, b, c with $ac - b^2 = \Delta$ and $|2b| \leq a \leq c$, this yields a classification of binary quadratic forms with integral coefficients of given negative discriminant.

Lagrange's result also directly implies that each lattice Λ in two dimensions contains a nonzero vector x such that

- (3) $\|x\| \leq \left(\frac{4}{3}\right)^{1/4} \sqrt{\det \Lambda}$.

It is not difficult to see that here the factor $\left(\frac{4}{3}\right)^{1/4}$ is best possible.

Gauss and Seeber

In Art. 40 of his famous book *Disquisitiones Arithmeticae*, Gauss [1801] showed that a linear diophantine equation has a solution if and only if the g.c.d. of the coefficients divides the right-hand side.

The reduction problem in two variables was studied in Artt. 171–181 of the *Disquisitiones*, while Art. 213 contains a rudimentary version of the Hermite normal form for 2×2 -matrices. In Artt. 266–285 Gauss starts the study of the reduction of ternary quadratic forms (i.e. in three variables). He proved that for each positive definite 3×3 -matrix D there exists an integral vector $x \neq 0$ with $x^T D x \leq \frac{4}{3} \sqrt[3]{\det D}$. Hence each lattice Λ in three dimensions contains a nonzero vector x with

$$(4) \quad \|x\| \leq \left(\frac{4}{3}\right)^{1/2} (\det \Lambda)^{1/3}.$$

Moreover, Gauss showed that there are only finitely many equivalence classes of integral positive definite 3×3 -matrices of given determinant.

Gauss' study of ternary quadratic forms was continued by Seeber [1831], who modified Gauss' concept of reduced form, and showed that each lattice Λ in three dimensions has a basis b_1, b_2, b_3 with $\|b_1\| \cdot \|b_2\| \cdot \|b_3\| \leq 3 \det \Lambda$. In an anonymous recension in the *Göttingische gelehrte Anzeigen*, Gauss [1831] simplified Seeber's arguments (however remarking: 'Wenn ein schwieriges Problem oder Theorem aufzulösen oder zu beweisen vorliegt, so ist allezeit der erste und mit gebührendem Danke zu erkennende Schritt, dass überhaupt eine Auflösung oder ein Beweis gefunden werde, und die Frage, ob dies nicht auf eine leichtere und einfachere Art hätte geschehen können, bleibt so lange eine müßige, als die Möglichkeit nicht zugleich durch die That entschieden wird.'). Moreover, Gauss reduced Seeber's factor 3 to the (best possible) 2. (Simple proofs of this were given by Dirichlet [1850] and Hermite [1850b].) Seeber [1831] also showed the existence of the Hermite normal form for 3×3 -matrices.

Dirichlet on diophantine approximation

Fundamental results in diophantine analysis were obtained in the 1840's and 1850's by Dirichlet and Hermite.

Dirichlet [1842] first remarks that it has been long known from the theory of continued fractions that for each irrational number x there exist infinitely many pairs of integers p, q with $|p - \alpha q| < 1/q$. Next with the help of the following principle:

$$(5) \quad \text{if } t+1 \text{ objects are put into } t \text{ boxes, there exists a box containing at least two objects}$$

(the pigeon-hole principle, or drawer principle, or Schubfachprinzip), he showed an extension:

$$(6) \quad \text{if } x_1, \dots, x_n, \varepsilon \in \mathbb{R} \text{ with } 0 < \varepsilon < 1, \text{ there exist integers } p_1, \dots, p_n, q \text{ such that } |x_1 p_1 + \dots + x_n p_n - q| < \varepsilon \text{ and } 1 \leq |p_i| \leq \varepsilon^{-1/n} \text{ for } i = 1, \dots, n.$$

Moreover, Dirichlet says that analogous assertions hold for two or more simultaneous equations.

Hermite

Hermite [1849] showed that if $x_1, \dots, x_n$ are relatively prime integers, then there exists an integral matrix A with $\det A = 1$ and with first row $(x_1, \dots, x_n)$.

In a series of letters to Jacobi, published in the *Journal für die reine und angewandte Mathematik*, Hermite [1850a] stated some basic results on quadratic forms, approximation and diophantine equations. In his first letter Hermite showed:

$$(7) \quad \text{for each positive definite matrix } D \text{ of order } n \text{ there exists a nonzero integral vector } x \text{ such that } x^T D x \leq \left(\frac{4}{3}\right)^{n-1/2} (\det D)^{1/n}.$$

This implies

- (8) *each lattice Λ in n dimensions contains a nonzero vector x with $\|x\| \leq (\frac{4}{3})^{(n-1)/4} (\det \Lambda)^{1/n}$.*

This extends the earlier results of Lagrange, Gauss, and Seeber for the cases $n = 2$ and 3 . Hermite's proof is based on an interesting induction argument, which in fact yields a (non-polynomial-time) algorithm for finding x as in (8): given $x_0 \in \Lambda \setminus \{0\}$, let $\Omega := \{y \in \mathbb{R}^n \mid y^\top x \in \mathbb{Z} \text{ for each } x \in \Lambda; y^\top x_0 = 0\}$. So Ω is a lattice in $n-1$ dimensions, and we can hence find $y_0 \in \Omega$ such that $\|y_0\| \leq (\frac{4}{3})^{(n-2)/4} (\det \Omega)^{1/(n-1)}$. Next find a nonzero vector x_1 in the lattice $K := \Lambda \cap \{x \mid y_0^\top x = 0\}$ with $\|x_1\| \leq (\frac{4}{3})^{(n-2)/4} (\det K)^{1/(n-1)}$ (which is again a problem in $n-1$ dimensions). It is not difficult to show $\det \Omega \leq \|x_0\|^{-1} (\det \Lambda)^{n-1}$ and $\det K \leq \|y_0\| \det \Lambda$, and hence:

$$(9) \quad \|x_1\| \leq (\frac{4}{3})^{(n^2-2n)(4n-4)/4} (\det \Lambda)^{(n-2)/(n-1)^2} \|x_0\|^{1/(n-1)^2}.$$

Now replace x_0 by x_1 , and start anew. (9) implies that by repeating this we finally find a vector x described in (8). (If $x_0 = x_1$ in (9) we obtain the inequality (8).) It is not difficult to see that this algorithm in fact finds a nonzero vector x in Λ with $\|x\| \leq (\frac{4}{3} + \epsilon)^{(n-1)/4} (\det \Lambda)^{1/n}$, in time which is, for each fixed n , bounded by a polynomial in the size of the given generator for Λ and in $1/\epsilon$.

Hermite mentions as an application of (8):

- (10) *if $\alpha_1, \alpha_2, \epsilon \in \mathbb{R}$ with $\epsilon > 0$, there exist integers p_1, p_2, q such that $|p_1 - \alpha_1 q| < \epsilon$, $|p_2 - \alpha_2 q| < \epsilon$, and $1 \leq q \leq \frac{8}{\epsilon} \sqrt{3} \cdot \epsilon^{-2}$.*

as follows from (8) by taking the lattice generated by

$$(11) \quad \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -\alpha_1 \\ -\alpha_2 \\ \frac{8}{\epsilon} \sqrt{3} \end{bmatrix}.$$

Extending a definition given by Gauss, Hermite defines (recursively) a positive definite $n \times n$ -matrix $D = (\delta_{ij})$ to be *reduced* if

- (12) (i) $\delta_{11} \leq (\frac{4}{3})^{(n-1)/2} (\det D)^{1/n}$ and $|\delta_{12}|, \dots, |\delta_{1n}| < \frac{1}{2} \delta_{11}$;
(ii) the positive definite $(n-1) \times (n-1)$ -matrix D' obtained from D^{-1} by deleting the first row and first column is reduced.

Hermite showed that each positive definite matrix is equivalent to a reduced matrix, and that, if we fix Δ , there are only finitely many reduced integral positive definite matrices with determinant Δ . A consequence is that there are only finitely many equivalence classes of integral positive definite matrices of given determinant, a result proved earlier by Eisenstein [1847].

In the second letter, Hermite showed:

- (13) *each positive definite $n \times n$ -matrix D is equivalent to a matrix whose diagonal entries have product at most $(\frac{4}{3})^{n(n-1)/4} \det D$.*

Equivalently:

- (14) *each full-dimensional lattice Λ in $\mathbb{R}^n$ has a basis $b_1, \dots, b_n$ such that $\|b_1\| \cdots \|b_n\| \leq (\frac{4}{3})^{n(n-1)/4} \det \Lambda$.*

In the third letter, Hermite states without proof:

- (15) *in each full-dimensional lattice Λ in $\mathbb{R}^n$, any basis $b_1, \dots, b_n$ for which $(\|b_1\|, \dots, \|b_n\|)$ is lexicographically minimal satisfies $\|b_1\| \cdots \|b_n\| \leq c(n) \det \Lambda$, where $c(n)$ only depends on n .*

(For a proof, see Stouff [1902].)

Hermite [1851] states as an auxiliary remark, without proof, that each rational matrix can be transformed by elementary column operations to a form which is now known as the Hermite normal form. The main topic of the paper is again the reduction of quadratic forms.

As we saw in Corollary 4.1a, the Hermite normal form implies:

- (16) *A rational system $Ax = b$ of linear equations has an integral solution x if and only if yb is an integer for each rational row vector y with yA integral.*

Heger and Smith on systems of linear diophantine equations

Heger [1856] showed that a system of *two* rational linear diophantine equations $Ax = b$ has a solution if and only if the g.c.d. of the 2×2 -subdeterminants of the matrix A divides the 2×2 -subdeterminants of the matrix $[A \ b]$. He also gave a formula for the general solution of two linear diophantine equations, mentioning that the process is essentially the same for systems with more equations. This is elaborated in Heger [1858], where also the above statement is extended to systems of m linear diophantine equations.

Smith [1861] gives a review on integral matrices and linear diophantine equations, and he derived several new results. He studies unimodular matrices (called *prime matrices*). Unknown of Heger's results, he showed Heger's characterization again. (In a footnote, Smith admits priority to Heger.) Moreover, Smith showed that each rational matrix can be brought into the form now called Smith normal form—see Section 4.4. Smith also proved that the set of integral solutions of a system $Ax = b$ of rational linear equations can be described as $\{x_0 + \lambda_1 x_1 + \dots + \lambda_t x_t \mid \lambda_1, \dots, \lambda_t \in \mathbb{Z}\}$, where $x_1, \dots, x_t$ are linearly independent and $t = (\text{number of variables}) - \text{rank}(A)$ (assuming $Ax = b$ has at least one integral solution). Smith's work was extended by Frobenius [1879, 1880].

Chebyshev, Kronecker, and Hurwitz on approximation

Chebyshev [1866] showed that if α and β are real numbers such that α/β is irrational, then there exist infinitely many pairs p, q of integers such that $|\alpha q - p - \beta| < 2/|q|$, which upper bound was reduced to $1/2|q|$ by Hermite [1880].

Kronecker elaborated the applications of Dirichlet's pigeon-hole principle (5) to approximation. Kronecker [1883] showed that if $Ax = b$ is a system of real linear equations, and $t > 0$, then $Ax = b$ can be satisfied with an approximation 'of the order of' $t^{-n/m}$ (m and n being the numbers of rows and columns of A), by a vector x having numerators and denominators 'of the same order as' t . His method implies that if A is an $m \times n$ -matrix with $m > n$, then for each $\varepsilon > 0$ there exists a nonzero integral vector x with $\|Ax\| < \varepsilon$. Another proof of this last statement is given in Kronecker [1884a]. In this paper also the elimination of variables and constraints in linear diophantine equations is studied.

Finally, Kronecker [1884b] combines many of the results above into the following famous theorem:

- (17) (Kronecker's approximation theorem) *Let A be a real $m \times n$ -matrix and let $b \in \mathbb{R}^m$. Then the following are equivalent:*
- (i) *for each $\varepsilon > 0$ there is an $x \in \mathbb{Z}^n$ with $\|Ax - b\| < \varepsilon$;*
 - (ii) *for each $y \in \mathbb{R}^m$, if $yA \in \mathbb{Z}^n$ then $yb \in \mathbb{Z}$.*

(Actually, Kronecker used the norm $\|\cdot\|_\infty$, but this is clearly irrelevant.) It is easy to see that (16) above is a corollary. (See also Perron [1921: Chapter V]. Several authors have re-proved (variants of) (17), e.g. Châtelet [1913: pp. 130–137; 1914], Giraud [1914], Kakeya [1913–4], and Riesz [1904].)

Hurwitz [1891] showed that for any irrational number α there exist infinitely many pairs of integers p, q with

$$(18) \quad \left| \alpha - \frac{p}{q} \right| < \frac{1}{\sqrt{5} \cdot q^2}$$

and that the factor $1/\sqrt{5}$ is best possible (as is shown by $\alpha = \frac{1}{2} + \frac{1}{2}\sqrt{5}$). In fact, infinitely many convergents p/q in the continued fraction method satisfy (18).

Poincaré [1880] studied the elementary properties of lattices in two dimensions (extended by Châtelet [1913] to more dimensions).

Minkowski's geometry of numbers

At the end of the 19th century, Minkowski found several extensions and sharpenings of the results above, by his *geometry of numbers*. The following result, presented in Minkowski [1893], is basic:

- (19) if $C \subseteq \mathbb{R}^n$ is a full-dimensional compact convex set, symmetric with respect to the origin, with volume at least 2^n , then C contains a nonzero integral vector.

(Symmetric with respect to the origin means $-x \in C$ if $x \in C$.) The proof is based on measuring volumes, as an extension of the pigeon-hole principle: suppose C contains no integral vector other than the origin. Then the sets $x + \frac{1}{2}C$, for $x \in \mathbb{Z}^n$, are pairwise disjoint. However:

$$(20) \quad \text{vol} \left(\bigcup_{x \in \mathbb{Z}^n} (x + \tfrac{1}{2}C) \cap [0, 1]^n \right) = \text{vol}(\tfrac{1}{2}C) \geq 1 = \text{vol}[0, 1]^n$$

which is a contradiction (as $[0, 1]^n$ is connected).

(19) is equivalent to:

- (21) for any norm $\|\cdot\|$ on $\mathbb{R}^n$, there exists a nonzero integral vector x such that $\|x\| \leq 2 \cdot U_n^{-1/n}$, where U_n is the volume of $\{y \mid \|y\| \leq 1\}$.

In particular, it implies a result of Minkowski [1891]:

- (22) for any positive definite matrix D there exists a nonzero integral vector x such that $x^T D x \leq 4 \cdot V_n^{-2/n} (\det D)^{1/n}$,

where V_n denotes the volume of the n -dimensional unit ball. As $4 \cdot V_n^{-2/n} \approx 2n/\pi e$, this sharpens Hermite's result (7) considerably.

[If we define γ_n as the smallest number which can replace the factor $4 \cdot V_n^{-2/n}$ in (22), then by Minkowski's result, $\gamma_n \leq 2n/\pi e$, while Minkowski [1893] showed $\gamma_n \geq n/2\pi e$ (here $\gamma_n \leq 2n/\pi e$ means $\limsup_{n \rightarrow \infty} (\gamma_n/n) \leq 2/\pi e$; similarly for $\geq$). So γ_n is of order linear in n . Blichfeldt [1914] showed that $\gamma_n \leq n/\pi e$, and recently Kabatyanskiĭ and Levenshtein [1978] that $\gamma_n \leq 0.872n/\pi e$. It is conjectured by several authors that $\lim_{n \rightarrow \infty} (\gamma_n/n) = 1/2\pi e$. At the moment the following values of γ_n are known exactly: $\gamma_1 = 1$, $\gamma_2 = 2/\sqrt{3}$, $\gamma_3 = \sqrt[3]{2}$, $\gamma_4 = \sqrt{2}$, $\gamma_5 = \sqrt[5]{8}$, $\gamma_6 = \sqrt[6]{64/3}$, $\gamma_7 = \sqrt[7]{64}$, $\gamma_8 = 2$ (as was shown by Korkine and Zolotareff [1872, 1873, 1877] and Blichfeldt [1935] (cf. Watson [1966])). See Chaundy [1946], Coxeter [1947], and Barnes [1959] for work on γ_9 and γ_{10} .]

In his famous book *Geometrie der Zahlen*, Minkowski [1896] gives a review of the above results and he derives some new theorems. In Section 37, he applies (21) to the norm $\|\cdot\|_\infty$:

- (23) for each nonsingular matrix A there exists a nonzero integral vector x such that $\|Ax\|_\infty \leq (\det A)^{1/n}$.

This *linear forms theorem* implies result (6) of Dirichlet.

Another corollary of (23) concerns simultaneous diophantine approximation, which was also considered by Dirichlet and Hermite (cf. 10)):

$$(24) \quad \text{if } x_1, \dots, x_n, \varepsilon \in \mathbb{R}, 0 < \varepsilon \leq 1, \text{ then there exist } p_1, \dots, p_n, q \in \mathbb{Z} \text{ with } 1 \leq q \leq \varepsilon^{-n} \text{ and } |x_i q - p_i| < \varepsilon \text{ for } i = 1, \dots, n$$

(Minkowski [1896: pp. 110–112]). Finally, Minkowski [1896: Sec. 50] showed:

$$(25) \quad \text{each lattice } \Lambda \text{ in } \mathbb{R}^n \text{ contains linearly independent vectors } b_1, \dots, b_n \text{ such that } \|b_1\| \cdots \|b_n\| \leq 2^n V_n^{-1} \det \Lambda.$$

This closes our historical review. Summarizing, we saw two main proof methods: iterative methods (Euclidean algorithm, continued fractions, Hermite's method), and packing methods (pigeon-hole principle, geometry of numbers). Roughly speaking, the latter methods give better bounds, but the first yield better algorithms, which have culminated in Lovász's *basis reduction method* (see Section 6.2 above).

FURTHER NOTES ON LATTICES AND LINEAR DIOPHANTINE EQUATIONS

Classical books on the subjects treated in this part are: for the reduction theory of quadratic forms: Dirichlet [1879], Bachmann [1898, 1923]; for the geometry of numbers: Minkowski [1896]; for continued fractions: Perron [1913] (including historical notes); for diophantine approximation: Minkowski [1907], Koksma [1936]; for diophantine equations: Skolem [1938].

Other and more recent books are: Bachmann [1902–1910] (Euclidean algorithm, continued fractions), Bergström [1935] (lattices, diophantine approximation), Cassels [1957] (diophantine approximation), Cassels [1959] (lattices, reduction, geometry of numbers), Châtelet [1913] (lattices, linear diophantine equations, reduction), Cohn [1962] (lattices, quadratic forms), Davenport [1952] (continued fractions, quadratic forms), Dickson [1929, 1939] (quadratic forms—introductory), Dickson [1930] (quadratic forms), Hancock [1939] (geometry of numbers, diophantine approximation), Hardy and Wright [1979] (Euclidean algorithm, continued fractions, diophantine approximation, geometry of numbers), Henriici [1977] (continued fractions), Hua [1982] (continued fractions, unimodular and integral matrices, geometry of numbers, quadratic forms), Jones and Thron [1980] (continued fractions), Khintchine [1956] (continued fractions), Knuth [1969: pp. 316–364] (Euclidean algorithm), Lang [1966b] (diophantine approximation), Lekkerkerker [1969] (lattices, reduction, geometry of numbers), Mordell [1969] (diophantine equations), Newman [1972] (integral matrices, Hermite and Smith normal form, introduction to lattices and geometry of numbers), Niven [1967] (continued fractions, diophantine approximation—introductory), Nonweiler [1984: Ch. 2] (continued fractions—numerical applications), Olds [1963] (continued fractions), Perron [1921] (diophantine approximation), Schmidt [1980] (diophantine approximation), Stark [1970] (Euclidean algorithm, continued fractions—introductory), Uspensky and Heaslet [1939: Ch. III] (Euclidean algorithm—introductory), and Wall [1948] (continued fractions). For a survey on algorithmic aspects, see Frumkin [1982]. Van der Waerden [1956] surveys the reduction theory of positive quadratic forms. Lagarias [1980b] studies the worst-case complexity of algorithms for problems like reducing a quadratic form.

Hammer [1977] collected unsolved problems concerning lattice points.

For historical accounts, see Dickson [1920: Ch. II] (linear diophantine equations), Dickson [1923] (quadratic forms), Smith [1925: pp. 418–421] (continued fractions), Jones and Thron [1980: pp. 1–11] (continued fractions), Scharlau and Opolka [1980], Kloyda [1937] (linear and quadratic equations 1550–1660), and Brezinski [1981] (continued fractions).